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Mahara

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(54) **MEASUREMENT DEVICE AND DISTANCE MEASURING DEVICE**

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G01S 17/08 (2006.01)

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CPC **G01S 7/486** (2013.01); **G01S 17/08** (2013.01)

(58) **Field of Classification Search**

USPC 356/5.01

See application file for complete search history.

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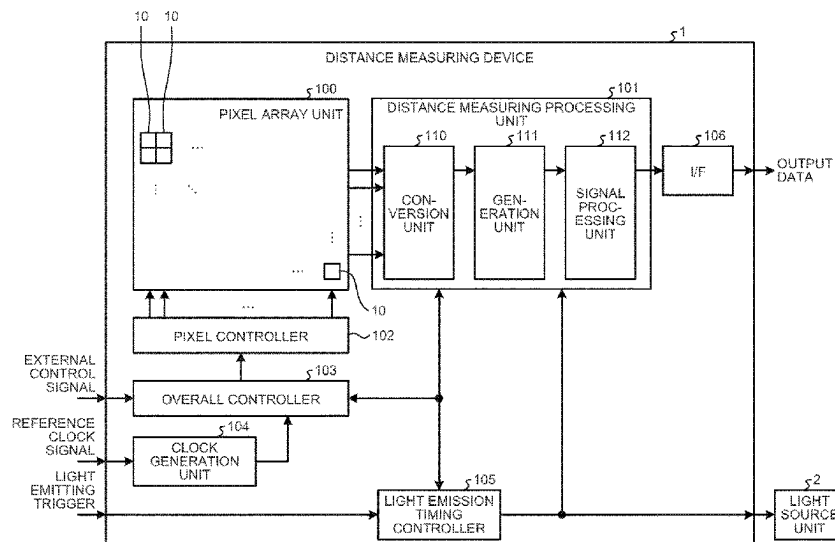
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(57) **ABSTRACT**

A measurement device includes: a light receiving unit including a plurality of pixels disposed in a target region in a matrix array, a controller that designates one or more target pixels, of a plurality of pixels, from which a signal is to be read, and a signal processing unit that performs a signal process based on a signal read from one or more target pixels included in a block having a predetermined width in the row direction of the array and having a variable height in the column direction. The controller switches, according to an instruction, between a first mode in which one or more target pixels are designated after performing thinning on the plurality of pixels, and a second mode in which one or more target pixels are designated without performing thinning.

18 Claims, 22 Drawing Sheets



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FIG.1

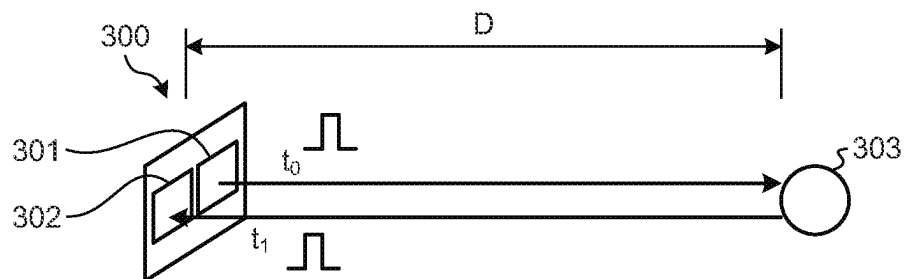


FIG.2

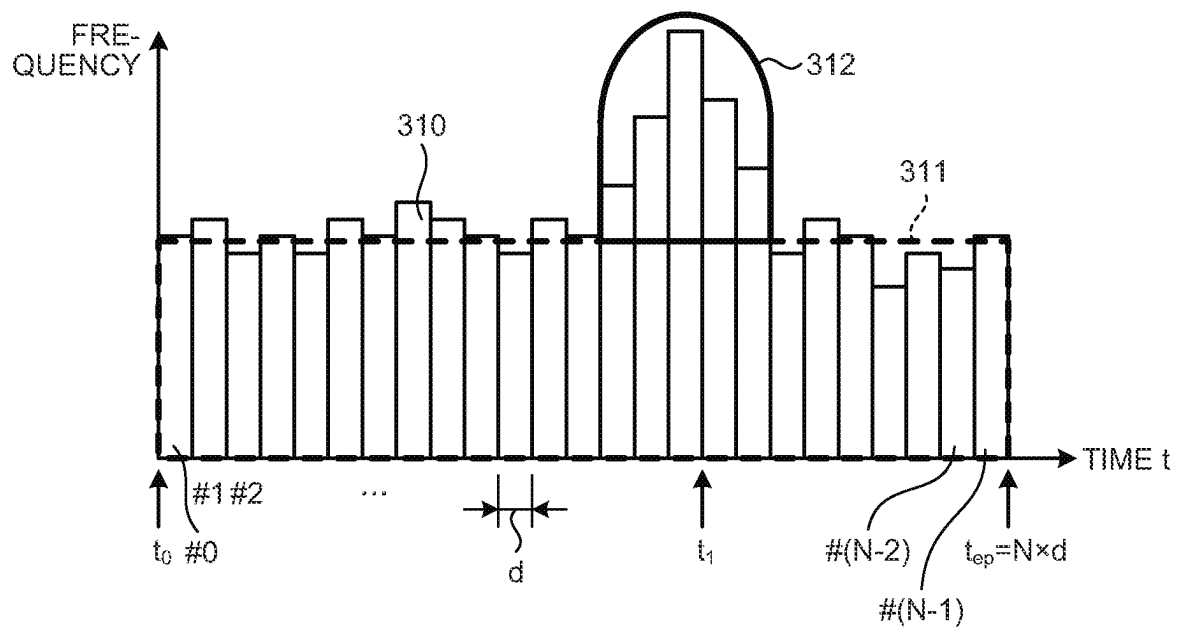


FIG.3

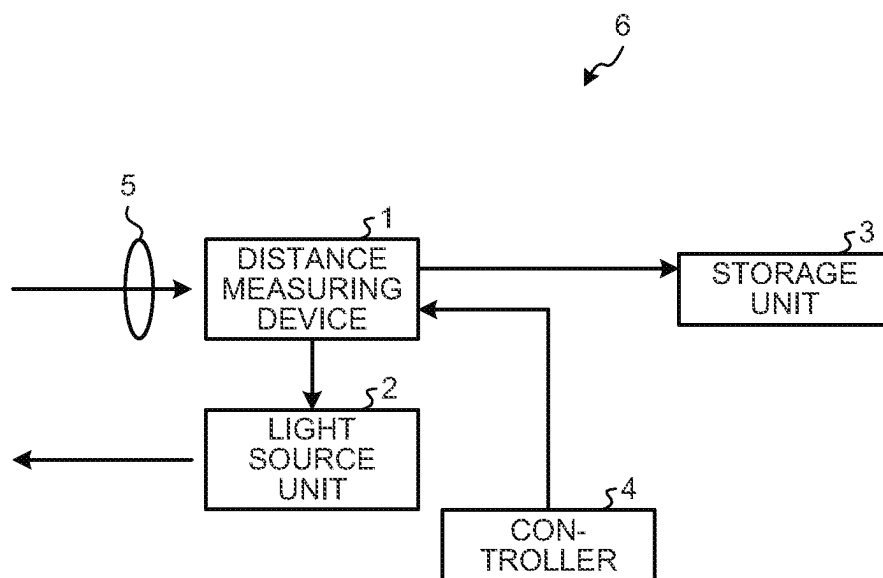


FIG.4

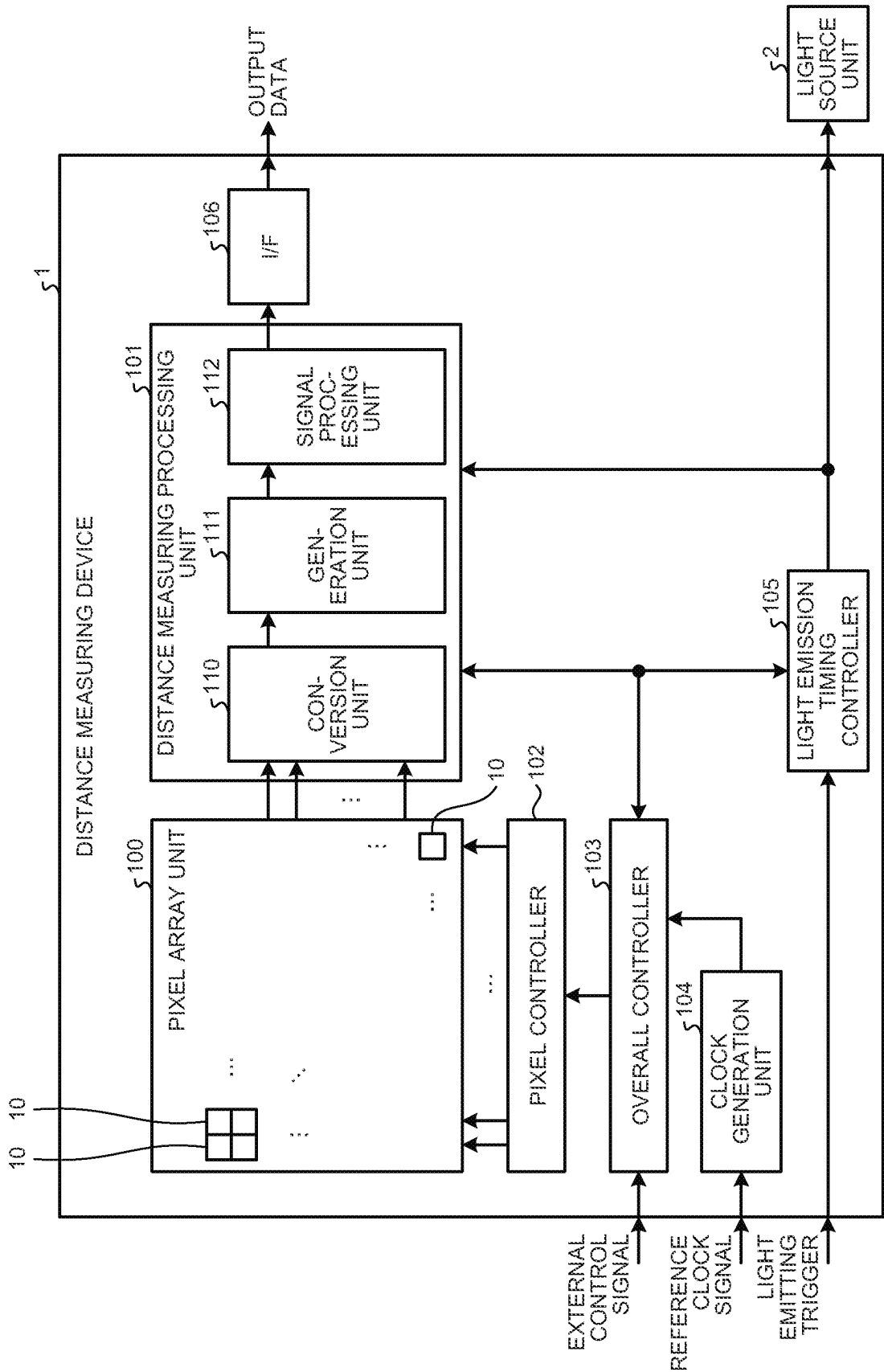


FIG.5

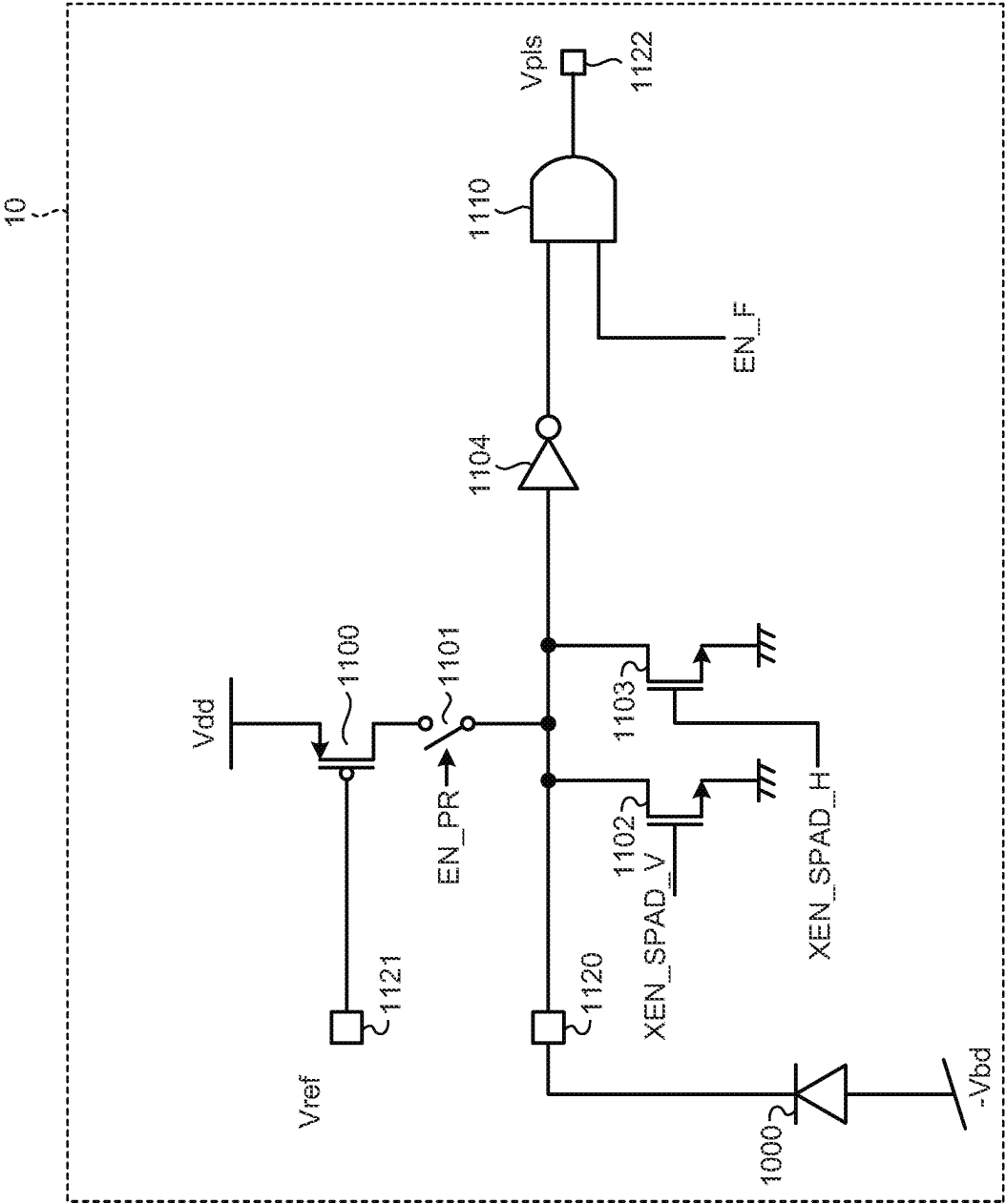


FIG.6

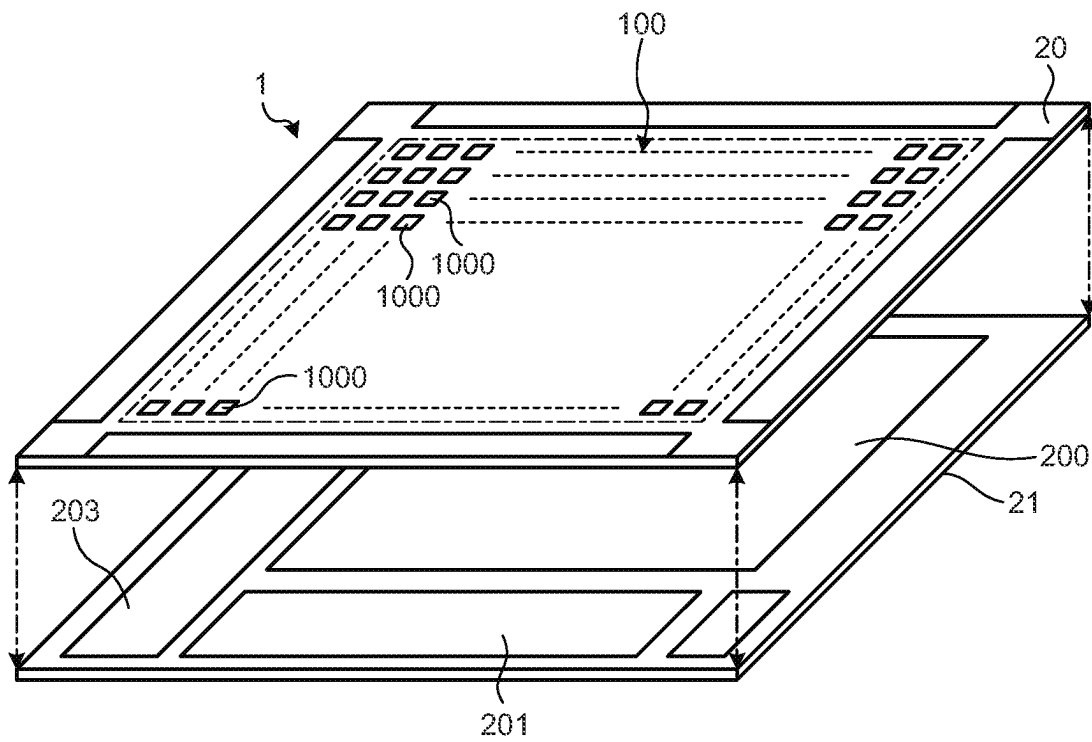


FIG. 7

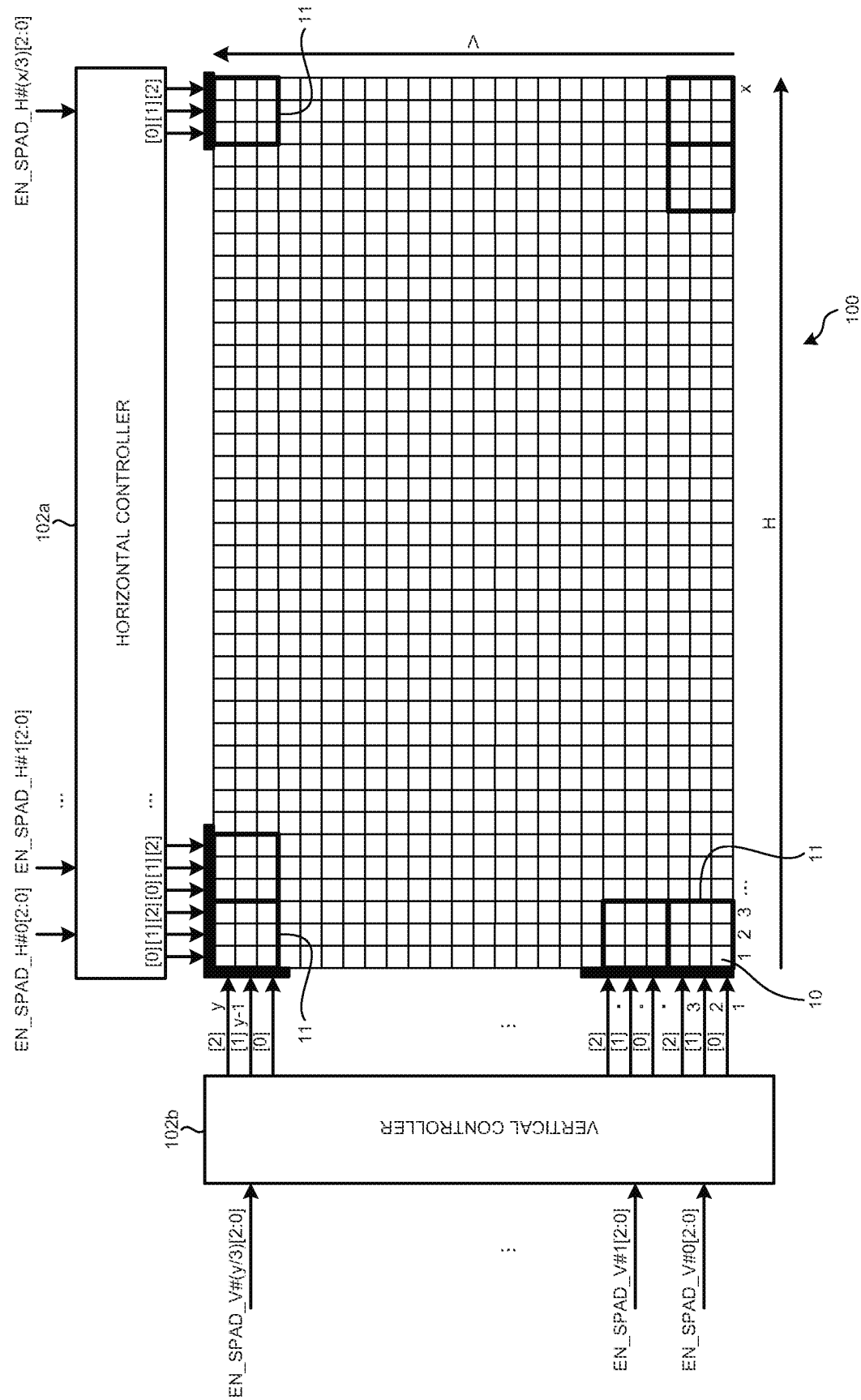


FIG.8A

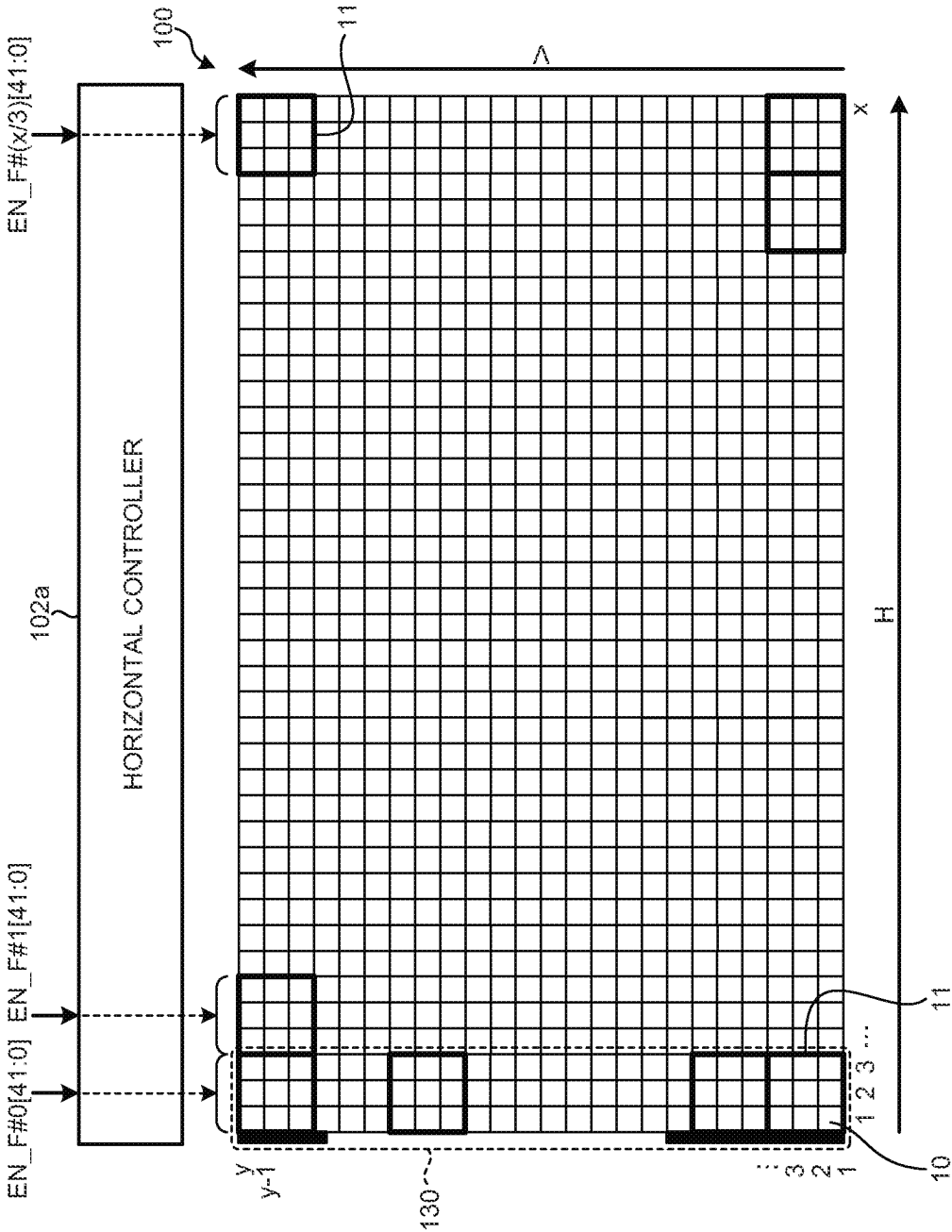


FIG.8B

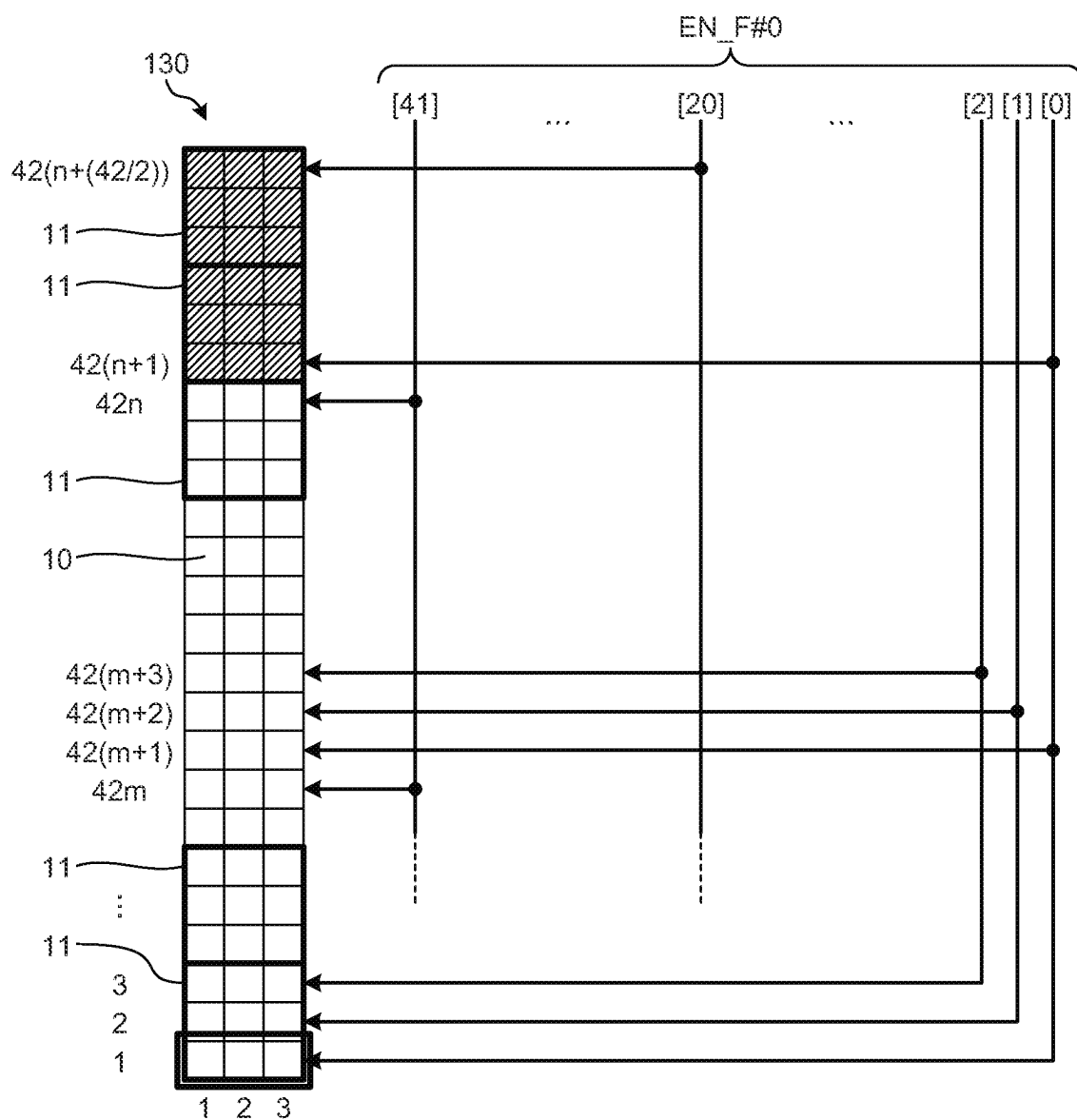


FIG.10

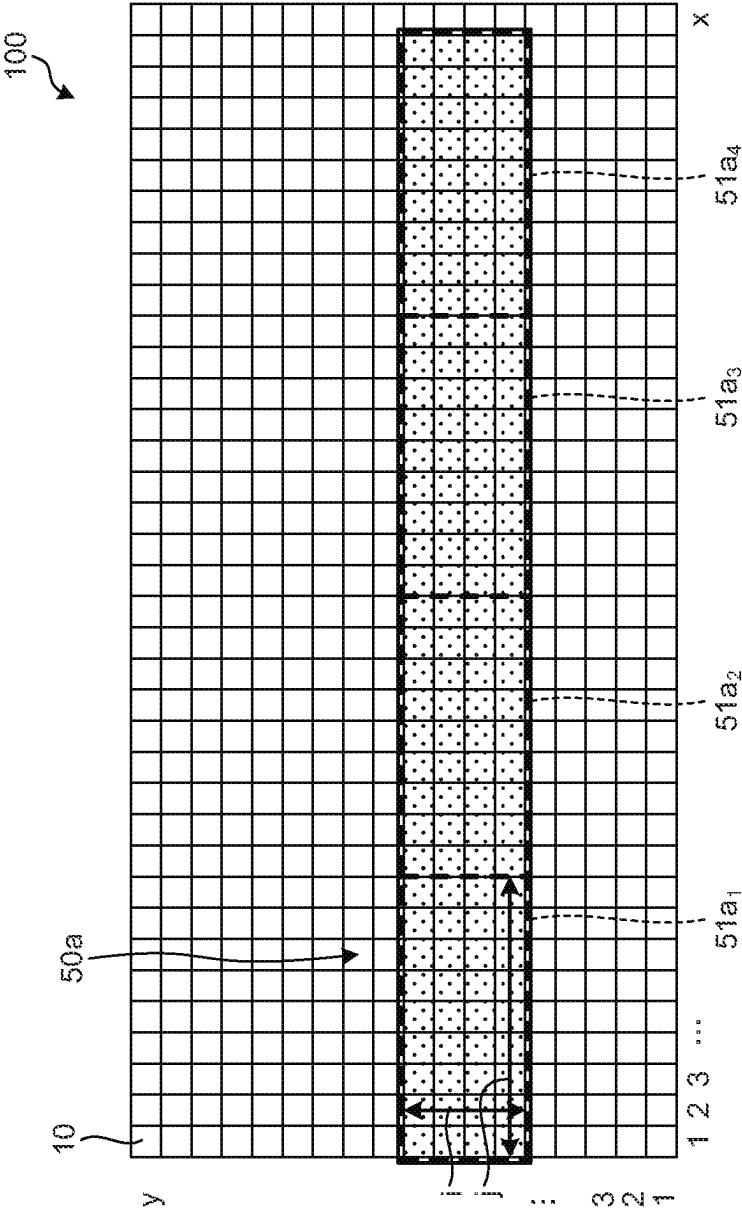


FIG.11

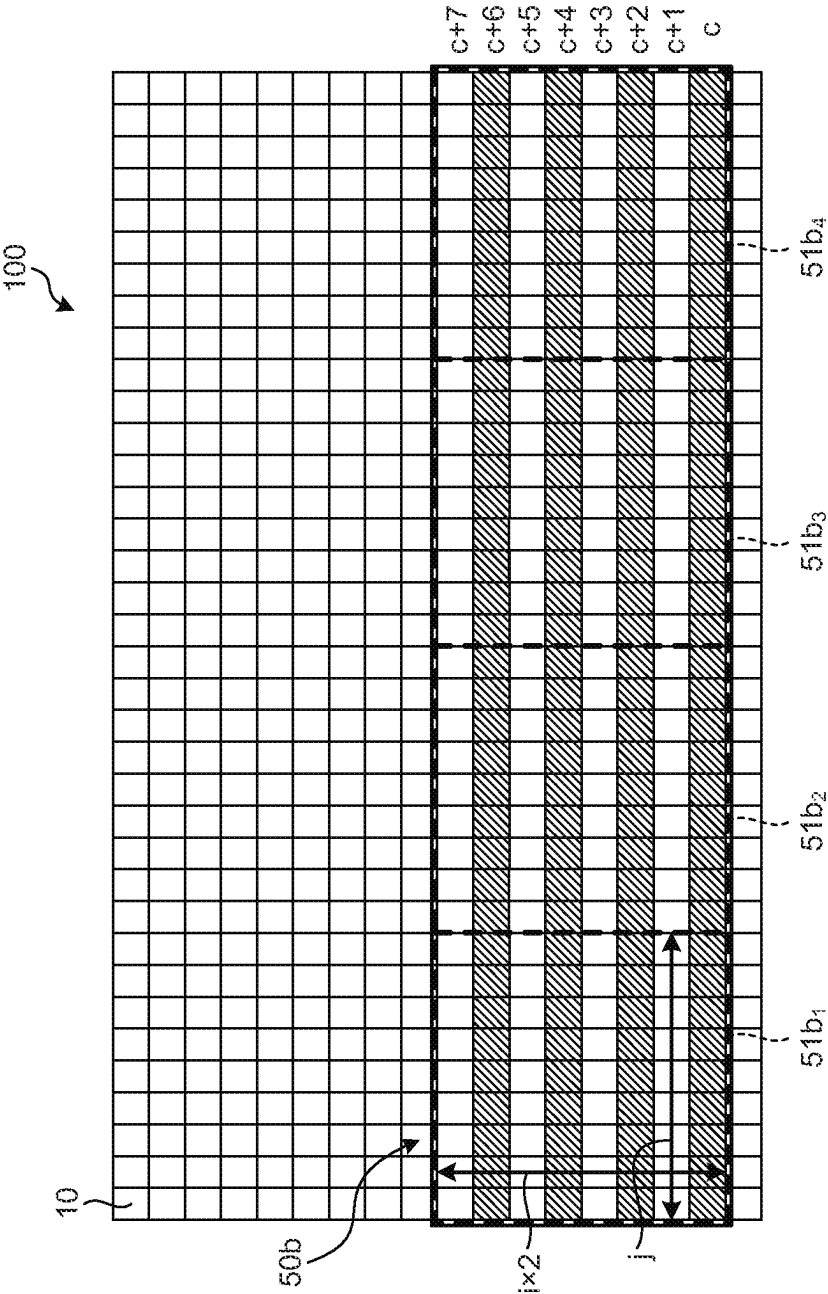


FIG.13

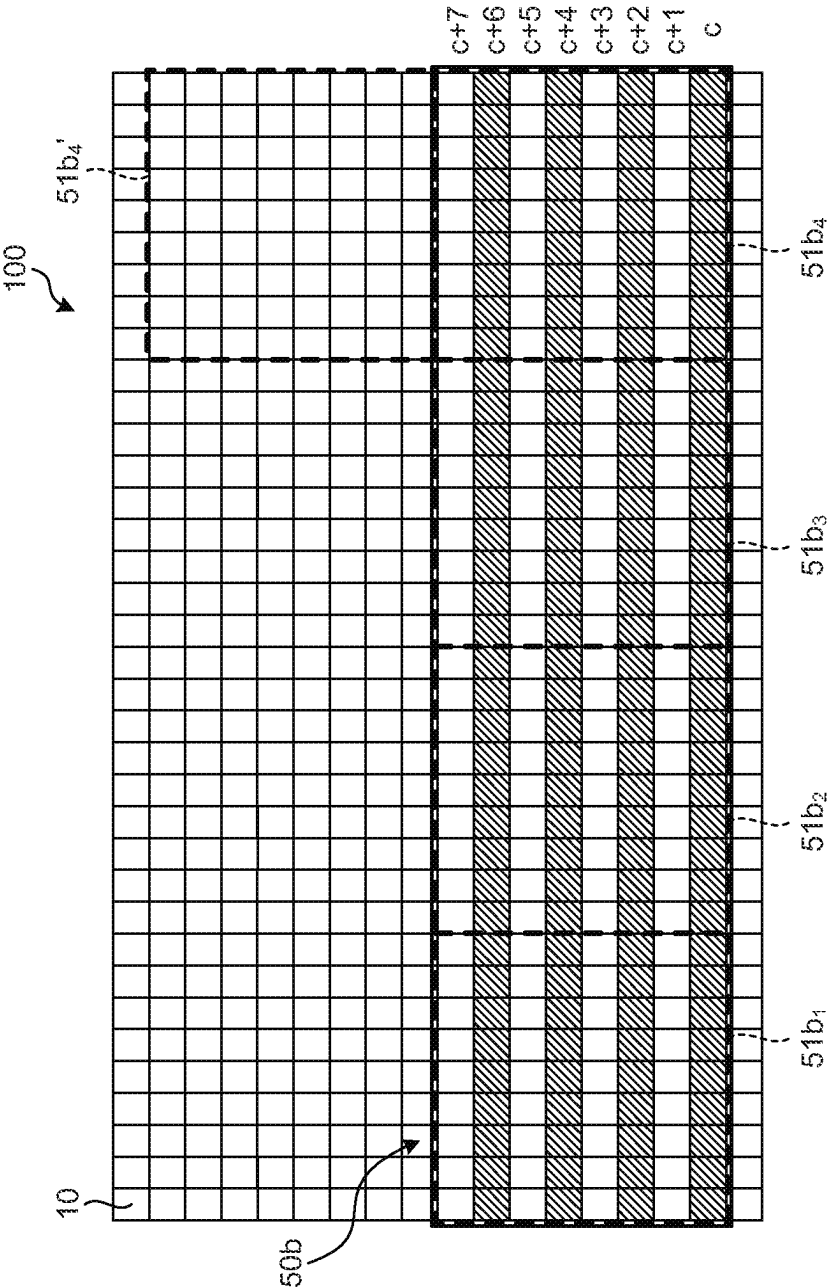


FIG. 14A

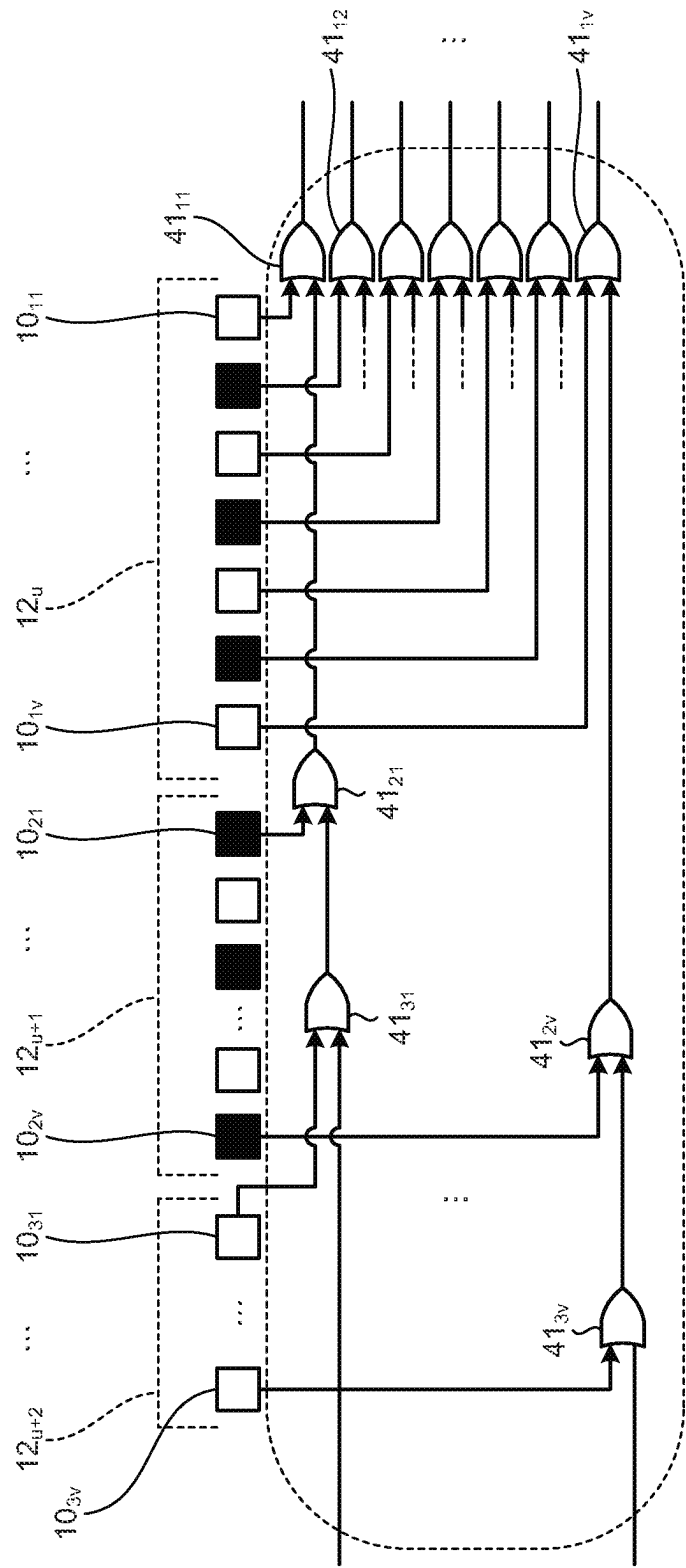


FIG.14B

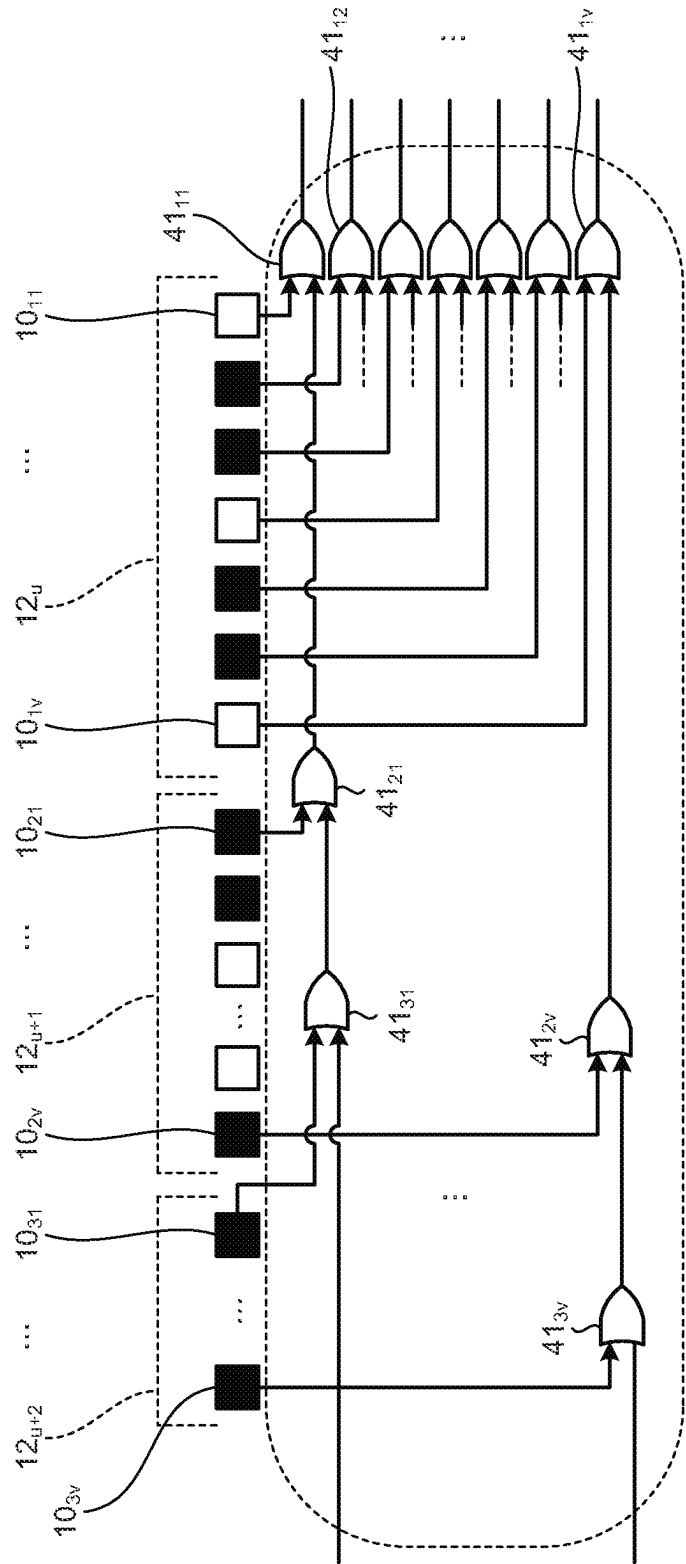


FIG.15

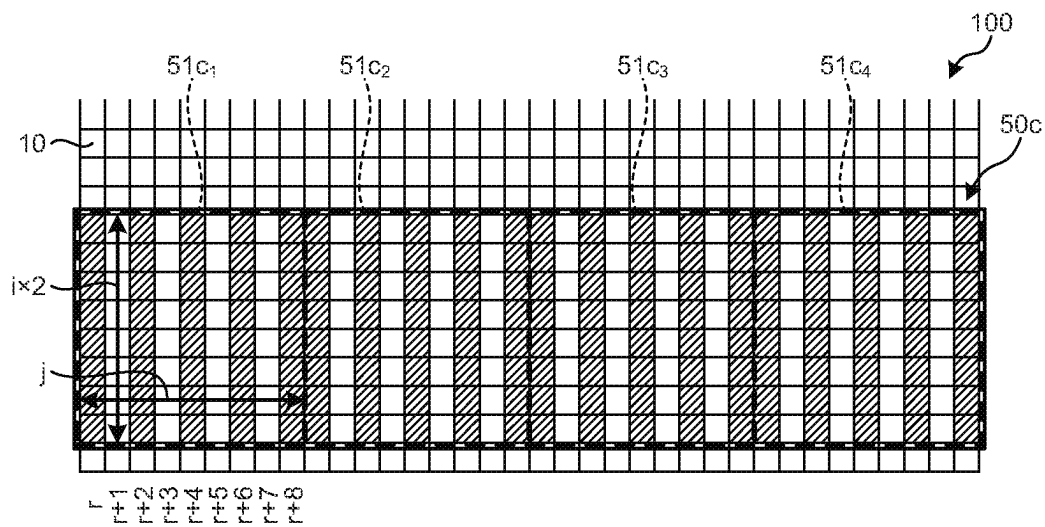


FIG.16

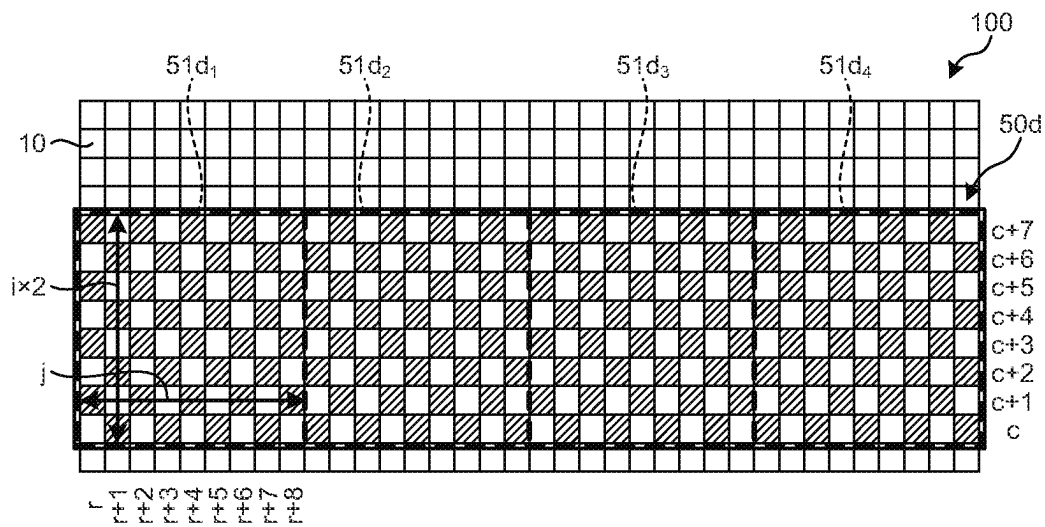


FIG.17

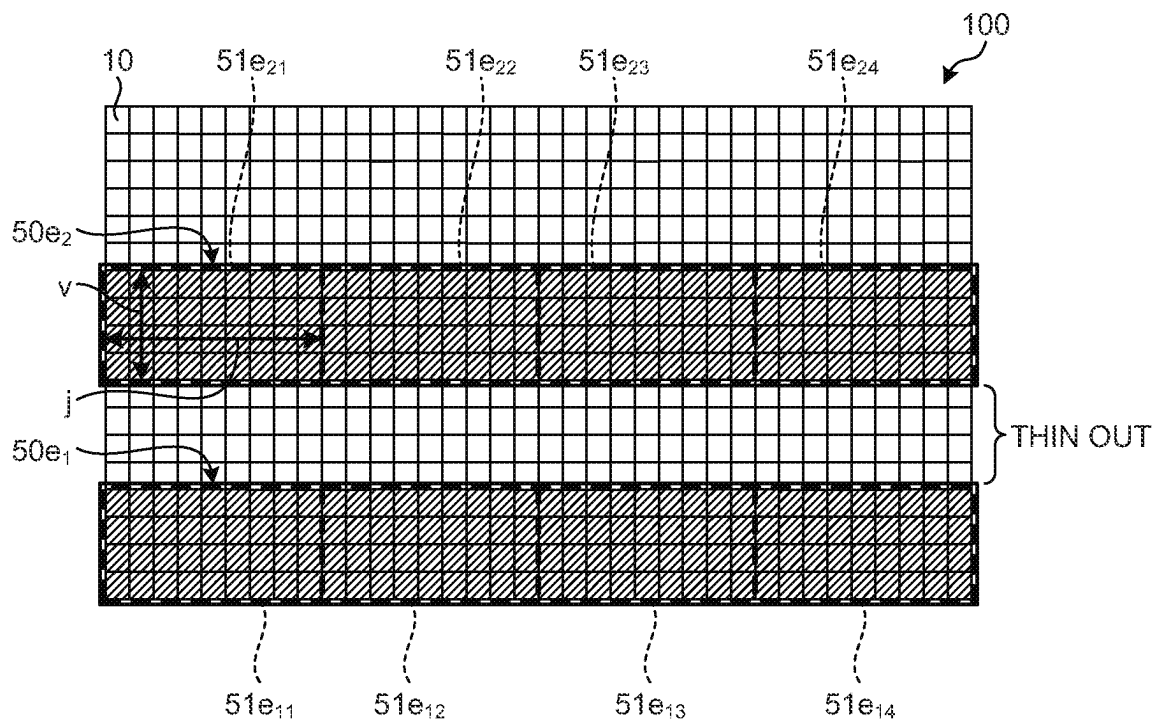


FIG.18

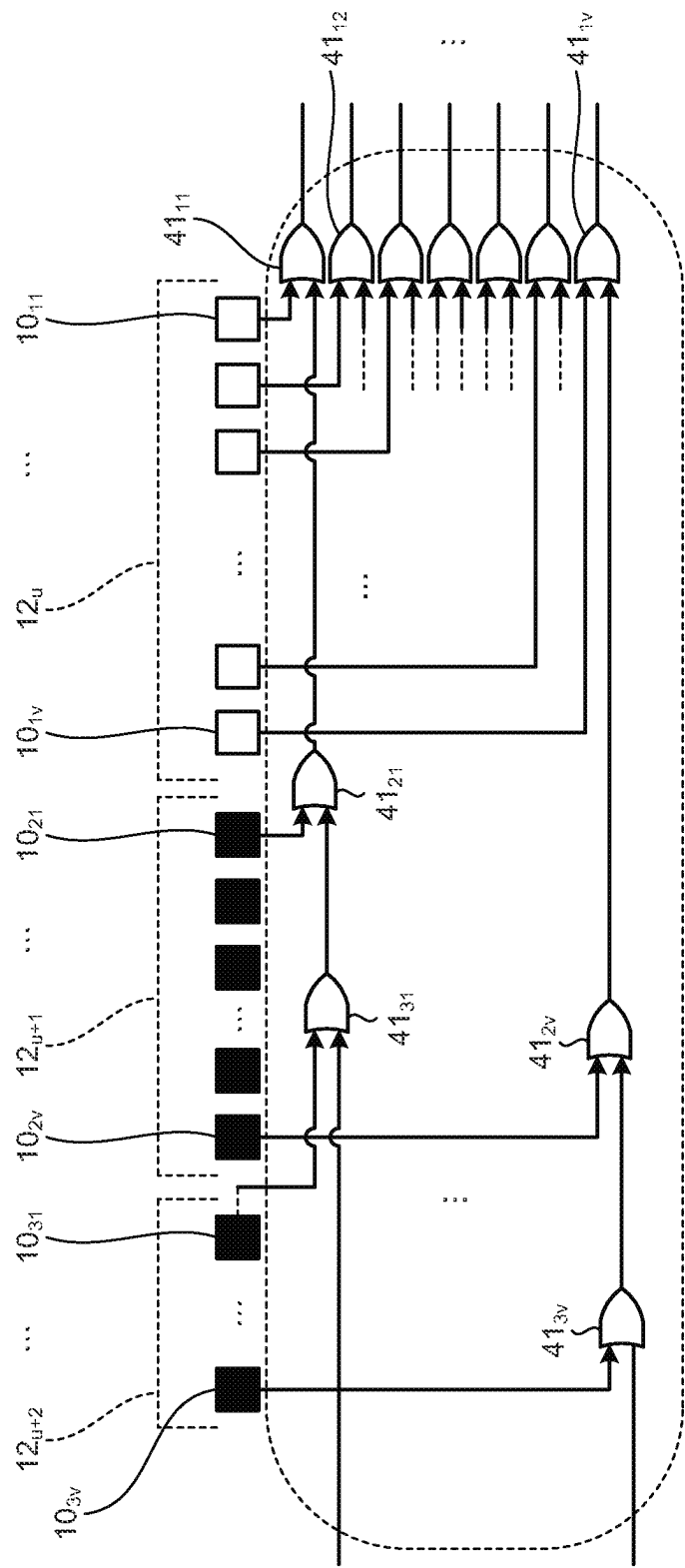


FIG.19

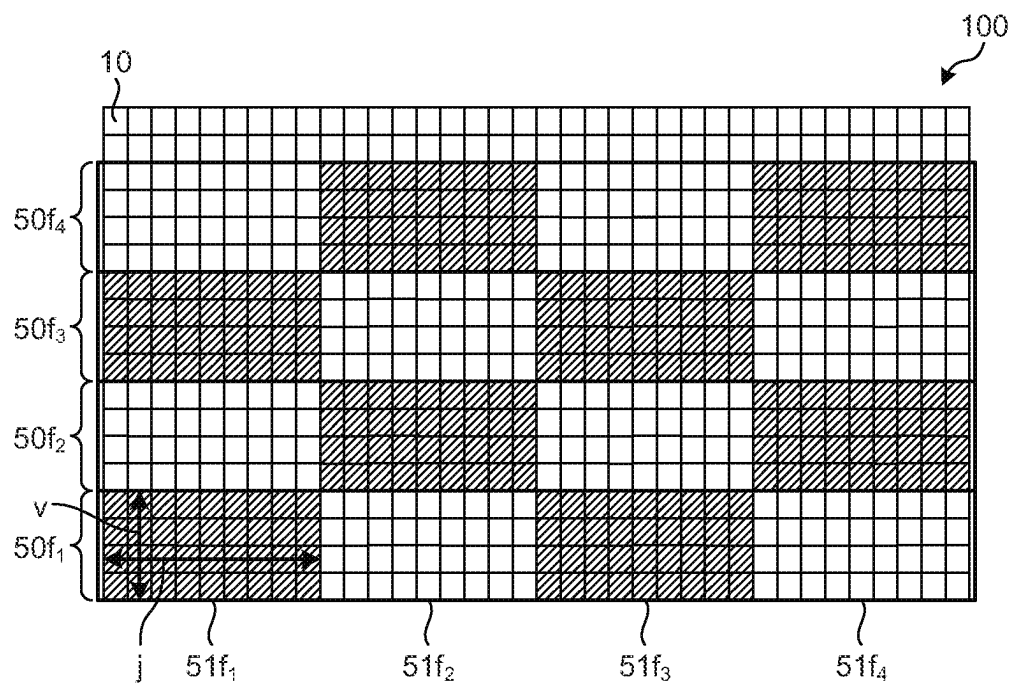


FIG.20

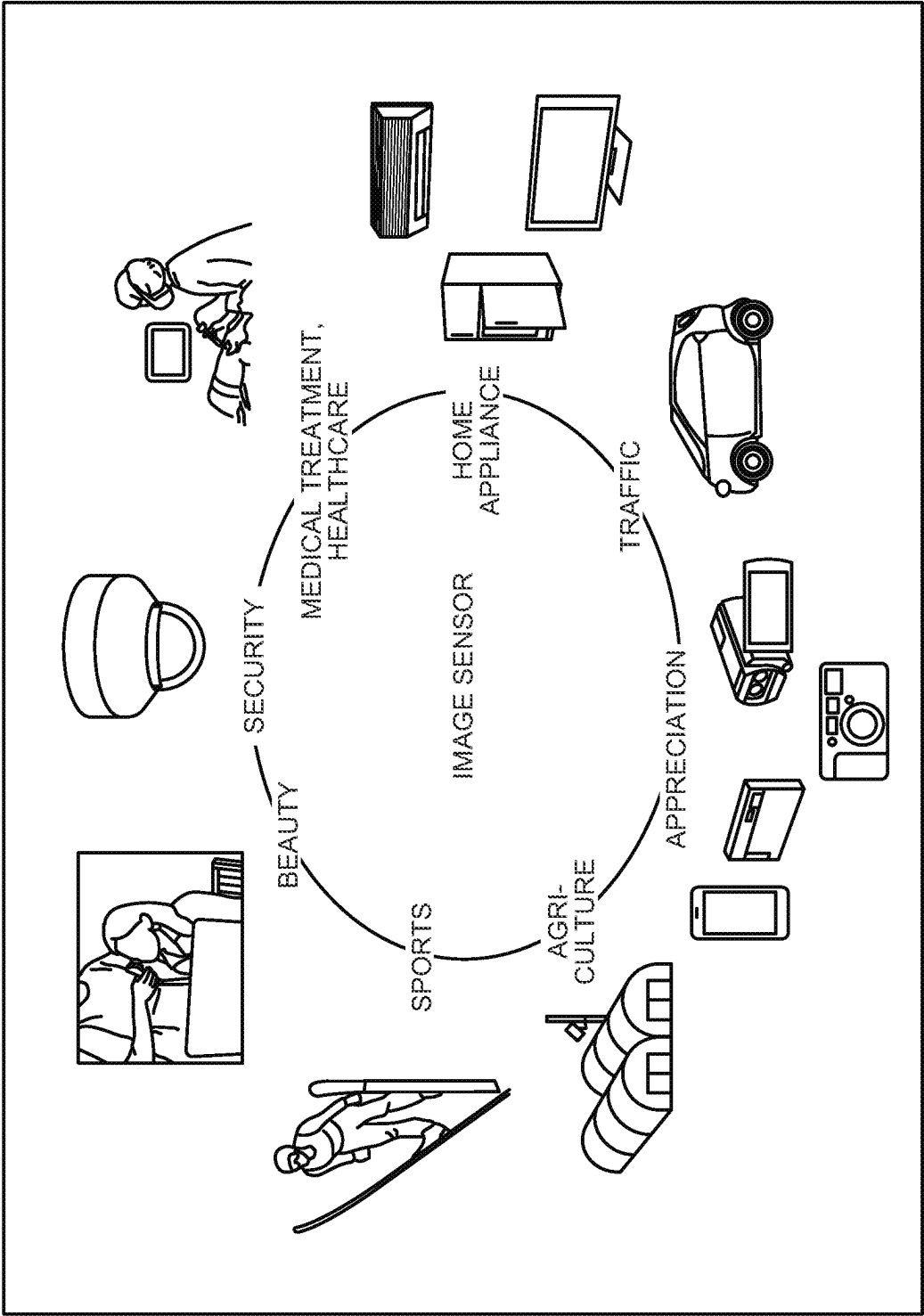


FIG.21

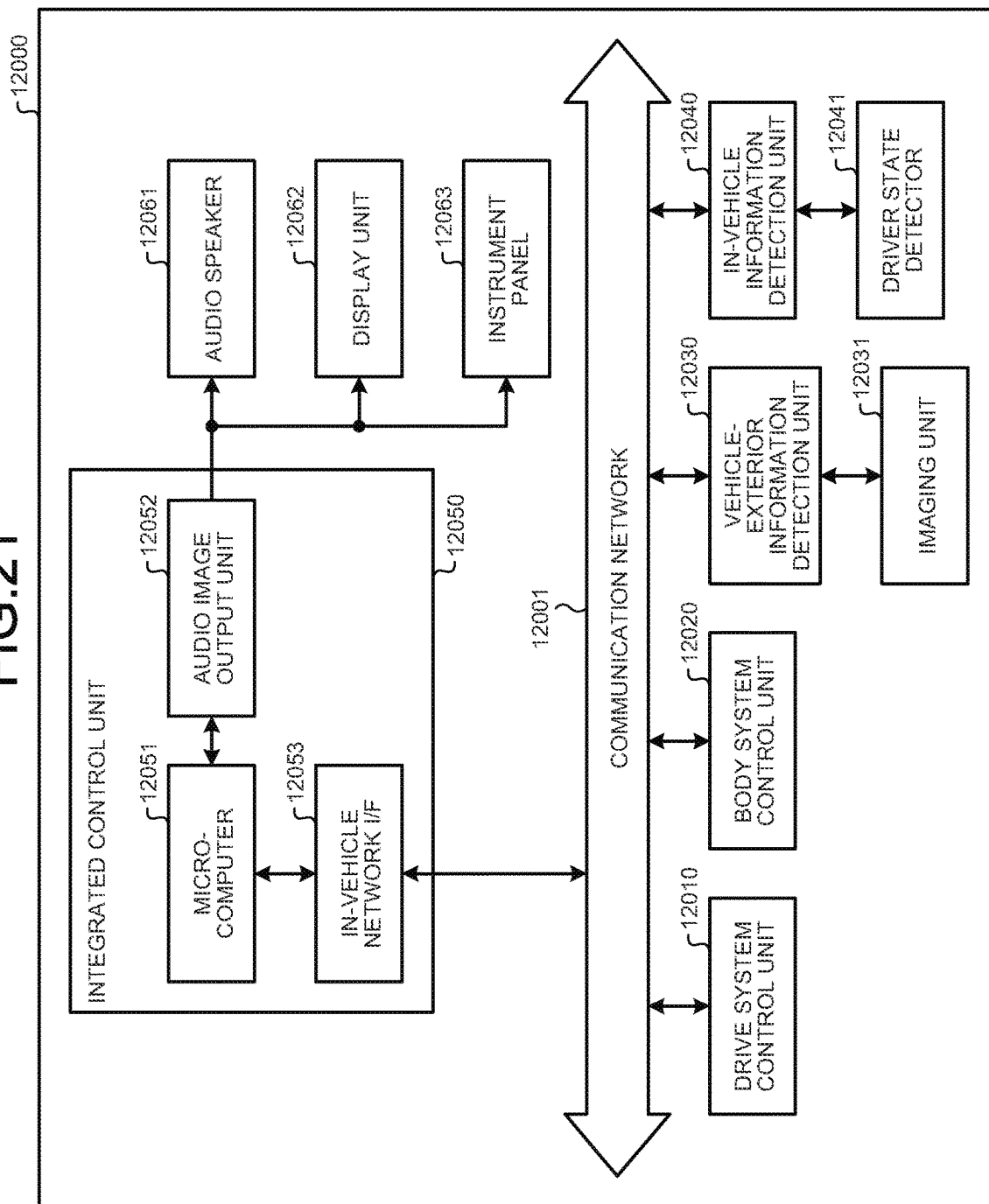
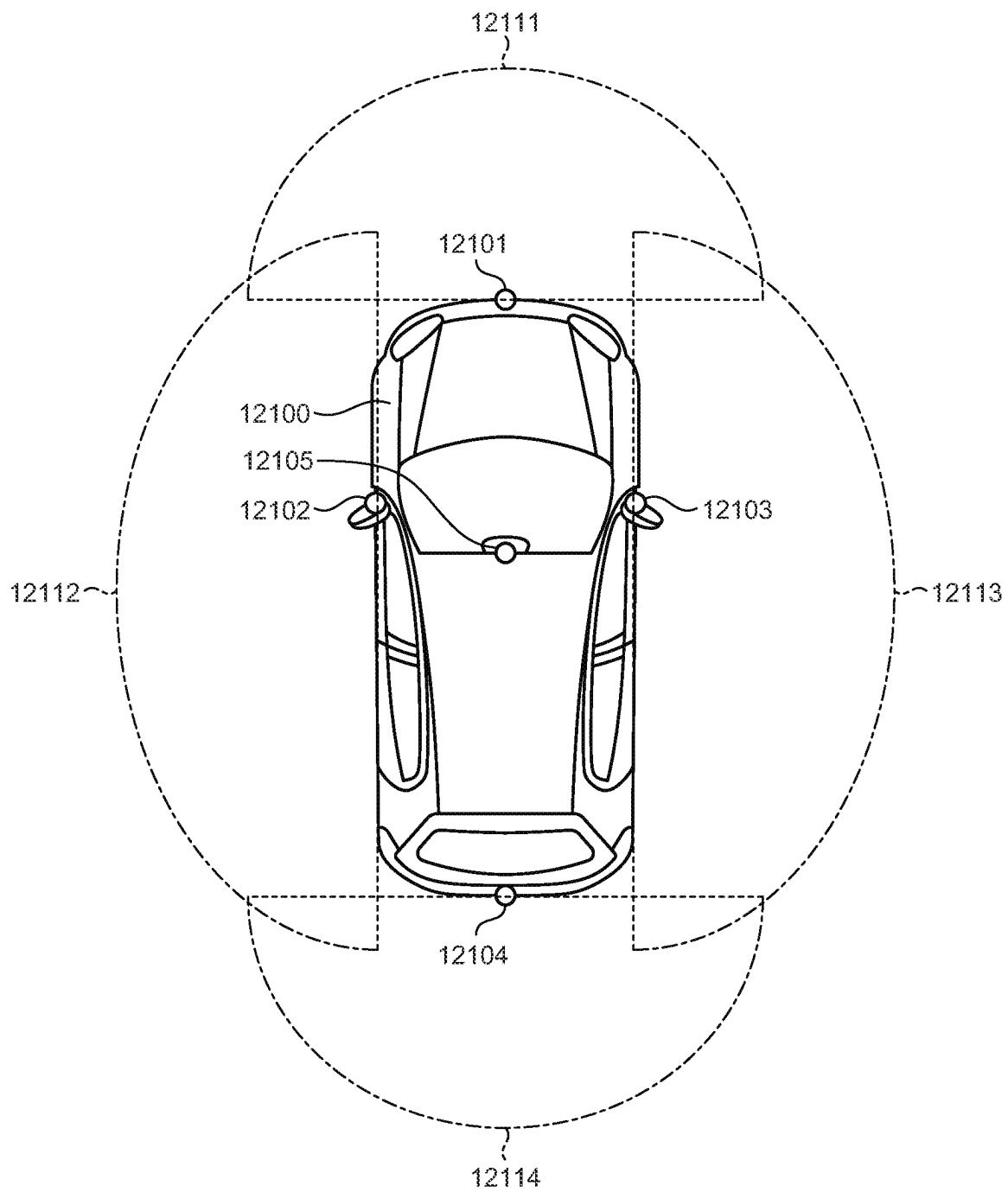


FIG.22



1

MEASUREMENT DEVICE AND DISTANCE MEASURING DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/001541 filed on Jan. 17, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-010673 filed in the Japan Patent Office on Jan. 24, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

FIELD

The present invention relates to a measurement device and a distance measuring device.

BACKGROUND

As one of the distance measuring methods for measuring the distance to an object to be measured using light, a distance measuring method called a direct time of flight (ToF) method is known. In the distance measuring process by the direct ToF method, the light receiving element receives the reflected light generated when the light emitted from the light source is reflected by the object to be measured. Based on the detection signal read from the light receiving element in response to the light reception of the reflected light, the time from the emission of the light to the reception as the reflected light is measured. A histogram is created based on the measured time, and the distance to the target is calculated based on this histogram. Further, in the direct ToF method, there is known a configuration in which distance measurement is performed using a pixel array in which light receiving elements are disposed in a two-dimensional lattice pattern.

CITATION LIST

Patent Literature

Patent Literature 1: JP 2018-044923 A

SUMMARY

Technical Problem

In the distance measurement using the pixel array described above, there is known a method of creating a histogram by simultaneously performing reading from a plurality of light receiving elements (during the same period). In this case, the region in which the reading of the plurality of light receiving elements can be performed at the same time is restricted in terms of the power consumption that drives the light receiving elements and the circuit scale. In order to mitigate this restriction, there is a demand for a technique capable of more efficiently reading the detection signal from the region.

An object of the present disclosure is to provide a measurement device and a distance measuring device capable of more efficiently reading a detection signal from a pixel array.

Solution to Problem

For solving the problem described above, a measurement device according to one aspect of the present disclosure has

2

a light receiving unit including a plurality of pixels disposed in a target region in a matrix array; a controller that designates one or more target pixels, of the plurality of pixels, from which a signal is to be read; and a signal processing unit that performs a signal process based on the signal read from the one or more target pixels included in a block having a predetermined width in a row direction of the array and having a variable height in a column direction, wherein the controller switches, according to an instruction, between a first mode in which the one or more target pixels are designated after performing thinning on the plurality of pixels, and a second mode in which the one or more target pixels are designated without performing the thinning.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram schematically illustrating distance measurement by the direct ToF method applicable to the embodiment.

FIG. 2 is a diagram illustrating an example histogram based on the time of light reception applicable to the embodiment.

FIG. 3 is a block diagram illustrating a configuration of an example of an electronic device including a distance measuring device according to the embodiment.

FIG. 4 is a block diagram illustrating in more detail the configuration of an example of a distance measuring device applicable to the embodiment.

FIG. 5 is a diagram illustrating a basic configuration example of a pixel circuit applicable to the embodiment.

FIG. 6 is a schematic diagram illustrating an example of the configuration of a device applicable to the distance measuring device according to the embodiment.

FIG. 7 is a diagram illustrating a more specific configuration example of a pixel array unit according to the embodiment.

FIG. 8A is a diagram illustrating an example of a detailed configuration of the pixel array unit according to the embodiment.

FIG. 8B is a diagram illustrating an example of a detailed configuration of the pixel array unit according to the embodiment.

FIG. 9 is a diagram illustrating an example of a configuration for reading a signal V_{pl}s from each pixel circuit according to the embodiment.

FIG. 10 is a diagram for explaining a method of scanning a pixel array unit by an existing technique.

FIG. 11 is a diagram for explaining a method of scanning a pixel array unit according to the first embodiment.

FIG. 12 is a diagram illustrating a comparison between a target pixel designated by the scanning method according to the first embodiment and a target pixel designated by the scanning method according to an existing technique.

FIG. 13 is a diagram for explaining mitigation of the influence from adjacent blocks by thinning out target pixels according to the first embodiment.

FIG. 14A is a diagram for explaining thinning corresponding to sharing of read lines according to the first embodiment.

FIG. 14B is a diagram for explaining thinning corresponding to sharing of read lines according to the first embodiment.

FIG. 15 is a diagram schematically illustrating an example of thinning according to the first modification of the first embodiment.

3

FIG. 16 is a diagram schematically illustrating an example of thinning according to the second modification of the first embodiment.

FIG. 17 is a diagram for explaining a method of scanning a pixel array unit according to the second embodiment.

FIG. 18 is a diagram for explaining thinning corresponding to sharing of read lines according to the second embodiment.

FIG. 19 is a diagram schematically illustrating an example of thinning according to the modification of the second embodiment.

FIG. 20 is a diagram illustrating a usage example using a distance measuring device according to the first embodiment and each modification thereof, and the second embodiment and the modification thereof according to the third embodiment.

FIG. 21 is a block diagram illustrating a schematic configuration example of a vehicle control system, which is an example of a moving object control system to which the technique according to the present disclosure can be applied.

FIG. 22 is a diagram illustrating an example of an installation position of an imaging unit.

DESCRIPTION OF EMBODIMENTS

Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the drawings. In the following embodiments, the same parts are designated by the same reference numerals, so that duplicate description will be omitted.

Technology Applicable to Each Embodiment

The present disclosure relates to a technique for performing distance measurement using light. Prior to the description of each embodiment of the present disclosure, the techniques applicable to each embodiment will be described for ease of understanding. In each embodiment, the direct time of flight (ToF) method is applied as the distance measuring method. The direct ToF method is a method in which the light receiving element receives the reflected light when the light emitted from the light source is reflected by the object to be measured, and the distance is measured based on the time difference between the light emission timing and the light reception timing.

The direct ToF method of distance measurement will be schematically described with reference to FIGS. 1 and 2. FIG. 1 is a diagram schematically illustrating distance measurement by the direct ToF method applicable to each embodiment. A distance measuring device 300 includes a light source unit 301 and a light receiving unit 302. The light source unit 301 is, for example, a laser diode, and is driven so as to emit the laser beam in a pulsed manner. The light emitted from the light source unit 301 is reflected by an object to be measured 303 and is received by the light receiving unit 302 as reflected light. The light receiving unit 302 includes a light receiving element that converts light into an electrical signal by photoelectric conversion to output a signal corresponding to the received light.

Here, the time when the light source unit 301 emits light (light emission timing) is time t_0 , and the time when the light receiving unit 302 receives the reflected light when the light emitted from the light source unit 301 is reflected by the object to be measured 303 (light reception timing) is t_1 . Assuming that the constant c is the speed of light (2.9979×10^8 [m/sec]), the distance D between the distance measuring

4

device 300 and the object to be measured 303 is calculated by the following equation (1).

$$D = (c/2) \times (t_1 - t_0) \quad (1)$$

The distance measuring device 300 repeats the above-mentioned process a plurality of times. The light receiving unit 302 may include a plurality of light receiving elements, and the distance D may be calculated based on each light reception timing when the reflected light is received by each light receiving element. The distance measuring device 300 classifies time t_m (called the light receiving time t_m) from time t_0 to the light emission timing to the light reception timing when the light is received by the light receiving unit 302 based on the class (bins) to generate a histogram.

The light received by the light receiving unit 302 during the light receiving time t_m is not limited to the reflected light when the light emitted from the light source unit 301 is reflected by the object to be measured. For example, the ambient light around the distance measuring device 300 (light receiving unit 302) is also received by the light receiving unit 302.

FIG. 2 is a diagram illustrating an example histogram based on the time when the light receiving unit 302 receives light applicable to each embodiment. In FIG. 2, the horizontal axis indicates the bin and the vertical axis indicates the frequency for each bin. The bin is a classification of the light receiving time t_m for each predetermined unit time d .

Specifically, bin #0 represents $0 \leq t_m < d$, bin #1 represents $d \leq t_m < 2d$, bin #2 represents $2d \leq t_m < 3d$, . . . , and bin #(N-2) represents $(N-2)d \leq t_m < (N-1)d$. When the exposure time of the light receiving unit 302 is time t_{ep} , $t_{ep} = Nd$.

The distance measuring device 300 counts the number of times the light receiving time t_m is acquired based on the bin to obtain a frequency 310 for each bin to generate a histogram. Here, the light receiving unit 302 also receives light other than the reflected light reflected when the light emitted from the light source unit 301 is reflected. As an example of such light other than the target reflected light, there is the above-mentioned ambient light. The portion indicated by a range 311 in the histogram includes the ambient light component due to the ambient light. The ambient light is light that is randomly incident on the light receiving unit 302 and is noise with respect to the reflected light of interest.

On the other hand, the target reflected light is light received according to a specific distance and appears as an active light component 312 in the histogram. The bin corresponding to the frequency of the peak in the active light component 312 is the bin corresponding to the distance D of the object to be measured 303. By acquiring the representative time of the bin (for example, the time in the center of the bin) as the time t_1 described above, the distance measuring device 300 can calculate the distance D to the object to be measured 303 according to the above equation (1). In this way, by using a plurality of light receiving results, it is possible to perform appropriate distance measurement for random noise.

FIG. 3 is a block diagram illustrating a configuration of an example of an electronic device including the distance measuring device according to each embodiment. In FIG. 3, an electronic device 6 includes a distance measuring device 1, a light source unit 2, a storage unit 3, a controller 4, and an optical system 5.

The light source unit 2 corresponds to the light source unit 301 described above and is a laser diode, and is driven so as to emit the laser beam in a pulsed manner, for example. A vertical cavity surface emitting laser (VCSEL) that emits the

5

laser beam can be applied for the light source unit **2** as a face light source. Not limited to this, the light source unit **2** may have a configuration in which an array in which laser diodes are disposed on a line may be used, and the laser beam emitted from the laser diode array is scanned in a direction perpendicular to the line. Furthermore, it may have a configuration in which a laser diode as a single light source is used and the laser beam emitted from the laser diode is scanned in the horizontal and vertical directions.

The distance measuring device **1** includes a plurality of light receiving elements corresponding to the light receiving unit **302** described above. The plurality of light receiving elements are, for example, disposed in a two-dimensional lattice to form a light receiving face. The optical system **5** guides light incident from the outside to the light receiving face included in the distance measuring device **1**.

The controller **4** controls the overall operation of the electronic device **6**. For example, the controller **4** supplies the distance measuring device **1** with a light emitting trigger that is a trigger for causing the light source unit **2** to emit light. The distance measuring device **1** causes the light source unit **2** to emit light at timing based on this light emitting trigger to store time to indicating the light emission timing. Further, the controller **4** sets a pattern for distance measurement for the distance measuring device **1** in response to an instruction from the outside, for example.

The distance measuring device **1** counts the number of times that time information (light receiving time t_m) indicating the timing at which light is received by the light receiving face is acquired within a predetermined time range to obtain the frequency for each bin to generate the above-mentioned histogram. The distance measuring device **1** further calculates the distance **D** to the object to be measured based on the generated histogram. The information indicating the calculated distance **D** is stored in the storage unit **3**.

FIG. **4** is a block diagram illustrating in more detail the configuration of an example of the distance measuring device **1** applicable to each embodiment. In FIG. **4**, the distance measuring device **1** includes a pixel array unit **100**, a distance measuring processing unit **101**, a pixel controller **102**, an overall controller **103**, a clock generation unit **104**, a light emission timing controller **105**, and an interface (I/F) **106**. The pixel array unit **100**, the distance measuring processing unit **101**, the pixel controller **102**, the overall controller **103**, the clock generation unit **104**, the light emission timing controller **105**, and the interface (I/F) **106** are disposed on, for example, one semiconductor chip.

In FIG. **4**, the overall controller **103** controls the overall operation of the distance measuring device **1** according to, for example, a program incorporated in advance. Further, the overall controller **103** can also perform control according to an external control signal supplied from the outside. The clock generation unit **104** generates one or more clock signals used in the distance measuring device **1** based on the reference clock signal supplied from the outside. The light emission timing controller **105** generates a light emission control signal indicating the light emission timing according to the light emitting trigger signal supplied from the outside. The light emission control signal is supplied to the light source unit **2** and to the distance measuring processing unit **101**.

The pixel array unit **100** includes a plurality of pixels **10**, each of which includes a light receiving element, which are disposed in a two-dimensional lattice pattern. The operation of each pixel **10** is controlled by the pixel controller **102** according to the instruction of the overall controller **103**. For example, the pixel controller **102** may

6

control the reading of the pixel signal from each pixel **10** for each block including (p×q) pixels **10** with p pixels in the row direction and q pixels in the column direction. Further, the pixel controller **102** can scan respective pixels **10** in the row direction and further scan them in the column direction with the block as a unit to read a pixel signal from each pixel **10**. Not limited to this, the pixel controller **102** can also control each pixel **10** independently. Further, the pixel controller **102** can set a predetermined region of the pixel array unit **100** as a target region, and can set the pixels **10** included in the target region as the target pixels **10** from which a pixel signal is to be read. Furthermore, the pixel controller **102** can collectively scan a plurality of rows (plural lines), further perform scanning in the column direction, and read a pixel signal from each pixel **10**.

In the following, scanning refers to a process in which the light source unit **2** (see FIG. **4**) is made to emit light, and the signal V_{pls} , corresponding to the light received, from the pixel **10** is continuously read for each pixels **10** designated as a scanning target in one scan region. It is possible to perform a plurality numbers of times of light emission and reading in one time scan.

The pixel signal read from each pixel **10** is supplied to the distance measuring processing unit **101**. The distance measuring processing unit **101** includes a conversion unit **110**, a generation unit **111**, and a signal processing unit **112**.

The pixel signal read from each pixel **10** to output from the pixel array unit **100** is supplied to the conversion unit **110**. Here, the pixel signal is asynchronously read from each pixel **10** and supplied to the conversion unit **110**. That is, the pixel signal is read from the light receiving element and is output according to the timing at which light is received by each pixel **10**.

The conversion unit **110** converts the pixel signal supplied from the pixel array unit **100** into digital information. That is, the pixel signal supplied from the pixel array unit **100** is output corresponding to the timing when light is received by the light receiving element included in the pixel **10** corresponding to the pixel signal. The conversion unit **110** converts the supplied pixel signal into time information indicating the timing.

The generation unit **111** generates a histogram based on the time information in which the pixel signal is converted by the conversion unit **110**. Here, the generation unit **111** counts the time information based on the unit time d set by a setting unit **113** to generate the histogram. The details of the histogram generation process by the generation unit **111** will be described later.

The signal processing unit **112** performs the predetermined arithmetic process based on the data of the histogram generated by the generation unit **111**, and calculates, for example, distance information. The signal processing unit **112** creates a curve approximation of the histogram based on, for example, the data of the histogram generated by the generation unit **111**. The signal processing unit **112** can detect the peak of the curve to which this histogram approximates and obtain the distance **D** based on the detected peak.

When the signal processing unit **112** performs a curve approximation of the histogram, the signal processing unit **112** can perform a filter process to the curve to which the histogram approximates. For example, the signal processing unit **112** can suppress a noise component by performing a low pass filter process on a curve to which the histogram approximates.

The distance information obtained by the signal processing unit **112** is supplied to an interface **106**. The interface **106** outputs the distance information supplied from the

signal processing unit **112** to the outside as output data. For example, a mobile industry processor interface (MIPI) can be used as the interface **106**.

In the above description, the distance information obtained by the signal processing unit **112** is output to the outside via the interface **106**, but the embodiment is not limited to this example. That is, the histogram data, which is data of the histogram generated by the generation unit **111**, may be output from the interface **106** to the outside. In this case, the information indicating the filter coefficient can be omitted from the distance measurement condition information set by the setting unit **113**. The histogram data output from the interface **106** is supplied to, for example, an external information processing device, and is appropriately processed.

FIG. **5** is a diagram illustrating a basic configuration example of the pixel **10** applicable to each embodiment. In FIG. **5**, the pixel **10** includes a light receiving element **1000**, transistors **1100**, **1102**, and **1103**, an inverter **1104**, a switch unit **1101**, and an AND circuit **1110**.

The light receiving element **1000** converts the incident light into an electrical signal by photoelectric conversion to output the signal. In each embodiment, the light receiving element **1000** converts the incident photon (photon) into an electrical signal by photoelectric conversion to output a pulse corresponding to the entry of the photon. In each embodiment, a single photon avalanche diode is used as the light receiving element **1000**. Hereinafter, the single photon avalanche diode is referred to as a single photon avalanche diode (SPAD). The SPAD has the characteristic in which when a large negative voltage that causes avalanche multiplication is applied to the cathode, the electrons generated in response to the entry of one photon cause avalanche multiplication and a large current flows. By utilizing this characteristic of the SPAD, it is possible to detect the entry of one photon with high sensitivity.

In FIG. **5**, the light receiving element **1000**, which is a SPAD, has a cathode connected to a coupling unit **1120** and an anode connected to a voltage source of the voltage ($-V_{bd}$). The voltage ($-V_{bd}$) is a large negative voltage that generates an avalanche multiplication for the SPAD. The coupling unit **1120** is connected to one end of the switch unit **1101** whose on (closed) and off (open) are controlled according to a signal EN_PR. The other end of the switch unit **1101** is connected to the drain of a transistor **1100**, which is a P-channel metal oxide semiconductor field effect transistor (MOSFET). The source of the transistor **1100** is connected to a power supply voltage Vdd. Further, a coupling unit **1121** to which a reference voltage Vref is supplied is connected to the gate of the transistor **1100**.

The transistor **1100** is a current source that outputs a current corresponding to the power supply voltage Vdd and the reference voltage Vref from the drain. With such a configuration, a reverse bias is applied to the light receiving element **1000**. When a photon is incident on the light receiving element **1000** with the switch unit **1101** turned on, the avalanche multiplication is started and a current flows from the cathode toward the anode of the light receiving element **1000**.

The signal extracted from the connection point between the drain of the transistor **1100** (one end of the switch unit **1101**) and the cathode of the light receiving element **1000** is input to the inverter **1104**. The inverter **1104** performs, for example, a threshold value determination on the input signal, inverts the signal each time the signal exceeds the

threshold value in the positive direction or the negative direction to output the signal as the signal Vpls, which is a pulsed output signal.

The signal Vpls output from the inverter **1104** is input to the first input port of the AND circuit **1110**. A signal EN_F is input to the second input port of the AND circuit **1110**. The AND circuit **1110** outputs the signal Vpls from the pixel **10** via a terminal **1122** when both the signal Vpls and the signal EN_F are in the high state.

In FIG. **5**, the coupling unit **1120** is further connected to the drains of transistors **1102** and **1103**, each of which is an N-channel MOSFET. The sources of the transistors **1102** and **1103** are connected, for example, to the ground potential. A signal XEN_SPAD_V is input to the gate of the transistor **1102**. A signal XEN_SPAD_H is input to the gate of the transistor **1103**. When at least one of these transistors **1102** and **1103** is in the off state, the cathode of the light receiving element **1000** is forcibly set to the ground potential, and the signal Vpls is set in the low state.

The signals XEN_SPAD_V and XEN_SPAD_H are used as vertical and horizontal control signals, respectively, in a two-dimensional lattice pattern in which respective pixels **10** are disposed in the pixel array unit **100**. As a result, the on/off state of each pixel **10** included in the pixel array unit **100** can be controlled for each pixel **10**. The on state of the pixel **10** is a state in which the signal Vpls can be output, and the off state of the pixel **10** is a state in which the signal Vpls cannot be output.

For example, in the pixel array unit **100**, the signal XEN_SPAD_H is set to the state in which the transistor **1103** is turned on for consecutive q columns of the two-dimensional lattice, and the signal XEN_SPAD_V is set to the state in which the transistor **1102** is turned on for consecutive p rows. As a result, the output of each light receiving element **1000** can be enabled in a block of p rows×q columns. In addition, since the signal Vpls is output by the AND circuit **1110** from the pixel **10** by the logical product with the signal EN_F, for example, it is possible to control whether the output of each light receiving element **1000** enabled by the signals XEN_SPAD_V and XEN_SPAD_H is enabled/disabled in more detail.

Further, by supplying the signal EN_PR that turns off the switch unit **1101**, for example, to the pixel **10** including the light receiving element **1000** whose output is to be disabled, it is possible to stop the supply of the power supply voltage Vdd to the light receiving element **1000**, and the pixel **10** can be turned off. This makes it possible to reduce the power consumption of the pixel array unit **100**.

These signals XEN_SPAD_V, XEN_SPAD_H, EN_PR, and EN_F are generated by the overall controller **103** based on the parameters stored in the register of the overall controller **103**, for example. The parameters may be stored in the register in advance, or may be stored in the register according to an external input. Each of the signals XEN_SPAD_V, XEN_SPAD_H, EN_PR, and EN_F generated by the overall controller **103** is supplied to the pixel array unit **100** by the pixel controller **102**.

The control by the signals EN_PR, XEN_SPAD_V, and XEN_SPAD_H using the switch unit **1101** and the transistors **1102** and **1103** described above is performed by using the analog voltage. On the other hand, the control by the signal EN_F using the AND circuit **1110** is performed by using the logic voltage. Therefore, the control by the signal EN_F can be performed at a lower voltage than the control by the signals EN_PR, XEN_SPAD_V, and XEN_SPAD_H, and is easy to handle.

FIG. 6 is a schematic diagram illustrating an example of a device configuration applicable to the distance measuring device 1 according to each embodiment. In FIG. 6, the distance measuring device 1 is configured by stacking a light receiving chip 20 made of a semiconductor chip, and a logic chip 21. In FIG. 5, for the sake of explanation, the light receiving chip 20 and the logic chip 21 are illustrated in a separated state.

In the light receiving chip 20, the light receiving elements 1000 included in the plurality of respective pixels 10 are disposed in a two-dimensional lattice pattern (matrix) in the region of the pixel array unit 100. Further, in the pixel 10, the transistors 1100, 1102, and 1103, the switch unit 1101, the inverter 1104, and the AND circuit 1110 are formed on the logic chip 21. The cathode of the light receiving element 1000 is connected via, for example, the coupling unit 1120 by a copper-copper connection (CCC) or the like between the light receiving chip 20 and the logic chip 21.

The logic chip 21 is provided with a logic array unit 200 including a signal processing unit that processes a signal acquired by the light receiving element 1000. The logic chip 21 can be further provided with a signal processing circuit unit 201 that processes the signal acquired by the light receiving element 1000, and an element controller 203 that controls the operation as the distance measuring device 1 in close proximity to the logic array unit 200.

For example, the signal processing circuit unit 201 can include the distance measuring processing unit 101 described above. Further, the element controller 203 can include the pixel controller 102, the overall controller 103, the clock generation unit 104, the light emission timing controller 105, and the interface 106 described above.

The configuration on the light receiving chip 20 and the logic chip 21 is not limited to this example. Further, the element controller 203 can be disposed for the purpose of the driving and control of another unit, for example, in the vicinity of the light receiving element 1000, in addition to the control of the logic array unit 200. Other than the arrangement illustrated in FIG. 6, the element controller 203 can be provided in an arbitrary region of the light receiving chip 20 and the logic chip 21 so as to have any functions.

FIG. 7 is a diagram illustrating a more specific configuration example of the pixel array unit 100 according to each embodiment. The pixel controller 102 described with reference to FIG. 4 is illustrated separately as a horizontal controller 102a and a vertical controller 102b in FIG. 7.

In FIG. 7, the pixel array unit 100 includes a total of (xxy) pixels 10 in x columns in the horizontal direction and y rows in the vertical direction. In addition, in FIG. 7 and the similar figures thereafter, the pixel 10 is indicated by the light receiving element 1000 included in the pixel 10 and having a rectangular light receiving face. That is, the pixel array unit 100 includes a configuration in which the light receiving faces of the light receiving elements 1000 as the pixels 10 are disposed in a matrix.

Further, in each embodiment, respective pixels 10 included in the pixel array unit 100 are controlled on a basis of the element 11 including a total of nine pixels 10, three in the horizontal direction and three in the vertical direction. For example, the signal EN_SPAD_H corresponding to the above-mentioned signal XEN_SPAD_H that controls respective pixels 10 in the row direction (horizontal direction), that is, on a column basis, is output from the overall controller 103 as a 3-bit signal (indicated as [2:0]) with the element 11 as a unit, and is supplied to the horizontal controller 102a. That is, by this one 3-bit signal, the signals EN_SPAD_H[0], EN_SPAD_H[1], and EN_SPAD_H[2] for

three pixels 10 disposed consecutively in the horizontal direction are merged and transmitted.

In the example of FIG. 7, the signals EN_SPAD_H #0 [2:0], EN_SPAD_H #1 [2:0], . . . , and EN_SPAD_H #(x/3) [2:0] are generated by the overall controller 103 in order from the element 11 at the left end of the pixel array unit 100, and are supplied to the horizontal controller 102a. The horizontal controller 102a controls each column of the corresponding element 11 according to the 3-bit value (indicated as [0], [1], [2]) of each signal EN_SPAD_H #0 [2:0], EN_SPAD_H #1 [2:0], . . . , and EN_SPAD_H #(x/3) [2:0].

Similarly, for example, the signal EN_SPAD_V corresponding to the above-mentioned signal XEN_SPAD_V that controls respective pixels 10 in the column direction (vertical direction), that is, on a row basis, is output from the overall controller 103 as a 3-bit signal with the element 11 as a unit, and is supplied to the vertical controller 102b. That is, by this one 3-bit signal, the signals EN_SPAD_V [0], EN_SPAD_V [1], and EN_SPAD_V [2] for three pixels 10 disposed consecutively in the vertical direction are merged and transmitted.

In the example of FIG. 7, the signals EN_SPAD_V #0 [2:0], EN_SPAD_V #1 [2:0], . . . , and EN_SPAD_V #(y/3) [2:0] are generated by the overall controller 103 in order from the element 11 at the lower end of the pixel array unit 100, and are supplied to the vertical controller 102b. The vertical controller 102b controls each row of the corresponding element 11 according to the 3-bit value of each signal EN_SPAD_V #0 [2:0], EN_SPAD_V #1 [2:0], . . . , and EN_SPAD_V #(y/3) [2:0].

Although not illustrated, for example, similar to the signal EN_SPAD_V above, the signal EN_PR is output from the overall controller 103 as a 3-bit signal with the element 11 as a unit, and is supplied to the vertical controller 102b. The vertical controller 102b controls each row of the corresponding element according to the 3-bit value of each signal EN_PR.

FIGS. 8A and 8B are diagrams illustrating an example of the detailed configuration of the pixel array unit 100 according to each embodiment. More specifically, FIGS. 8A and 8B illustrate control by the signal EN_F.

As illustrated in FIG. 8A, the signal EN_F is a signal supplied for a control target 130 including a plurality of adjacent columns of the pixel array unit 100. Here, the control target 130 is illustrated as including three columns that match the size of the element 11. Further, as the signal EN_F, the same signal is supplied to each row included in the control target 130 for each row with a predetermined cycle. That is, in this example in which the control target 130 includes three columns, the same signal EN_F is supplied to the three pixels 10 in the same row. In FIG. 8A, as an example, the signal EN_F is a 42-bit (indicated as [41:0]) signal, and is illustrated as the same signal being supplied every 42 rows (7 rows×6). In the example of FIG. 8A, the signals EN_F #0 [41:0], EN_F #1 [41:0], . . . , and EN_F #(x/3) [41:0] are output by the overall controller 103 from the left end of the pixel array unit 100 every three columns, and are supplied to the horizontal controller 102a.

The horizontal controller 102a supplies each bit of respective signals EN_F #0 [41:0], EN_F #1 [41:0], . . . , and EN_F #(x/3) [41:0] to each row of the corresponding control target 130. As illustrated in FIG. 8B, the horizontal controller 102a supplies the signal EN_F #0 [0] to, for example, to the control target 130 at the left end of the pixel array unit 100 every 42 rows, such as the first row, the 42 (m+1)th row (where m is an integer of one or more), . . . , the 42 (n+1)th

11

row, Similarly, the horizontal controller **102a** supplies the signal EN_F #0 [2] to the second row, the 42 (m+2)th row, . . . every 42 rows. In FIG. 8B, the signal EN_F #0 is supplied to the row of the control target **130** at the upper end, which corresponds to the first half of the unit of 42 rows.

That is, by this 42-bit signal EN_F[41:0], the signals EN_F[0], EN_F[1], . . . , and EN_F for 42 sets, of three pixels **10** disposed consecutively in the horizontal direction, which are disposed consecutively in the vertical direction are merged and transmitted.

In this way, it is possible to control the pixel array unit **100** differently for a plurality of columns by the signal EN_F. Further, the plurality of columns of the pixel array unit **100** is supplied with the same signal EN_F for a plurality of rows. Therefore, it is possible to control the respective pixels **10** included in the pixel array unit **100** with the plurality of rows as a cycle, with the plurality of rows as the minimum unit in the width direction.

FIG. 9 is a diagram illustrating an example of a configuration for reading the signal Vpls from each pixel **10** according to each embodiment. In FIG. 9, as indicated by the arrow in the figure, the horizontal direction of the figure is the column direction.

In each embodiment, a read line for reading the signal Vpls is shared for a predetermined number of pixels **10** in the column direction. In the example of FIG. 9, the read line is shared for the v pixels **10**. For example, consider groups $12_u, 12_{u+1}, 12_{u+2}, \dots$ each of which includes the v pixels **10** disposed in a row. The group 12_u includes the pixels 10_{11} to 10_{1v} , the group 12_{u+1} includes the pixels 10_{21} to 10_{2v} , and the group 12_{u+2} includes the pixels 10_{31} to $10_{3v}, \dots$

In the respective groups $12_u, 12_{u+1}, 12_{u+2}, \dots$, the read lines of the pixels **10** in which positions in the groups correspond are shared. In the example of FIG. 9, when the right side of the figure is regarded as the beginning of the position, the read lines of the first pixel 10_{11} of the group 12_u , the first pixel 10_{21} of the group 12_{u+1} , the first pixel 10_{31} of the group $12_{u+2}, \dots$ are shared. In the example of FIG. 9, a plurality of read lines of respective pixels $10_{11}, 10_{21}, 10_{31}, \dots$ are sequentially connected to each other via OR circuits $41_{11}, 41_{21}, 41_{31}, \dots$, respectively, to share the plurality of read lines.

For example, for the group 12_u , the pixels 10_{11} to 10_{1v} included in group 12_u are provided with respective OR circuits $41_{11}, 41_{12}, \dots$, and 41_{1v} , and the read lines of the pixels 10_{11} to 10_{1v} is connected to respective first input ports. Similarly for the group 12_{u+1} , the pixels 10_{21} to 10_{2v} included in the group 12_{u+1} are provided with OR circuits 41_{21} to 41_{2v} , respectively. Similarly for the group 12_{u+2} , the pixels 10_{31} to 10_{3v} included in the group 12_{u+2} are provided with OR circuits 41_{31} to 41_{3v} , respectively.

The output of each of the OR circuits 41_{11} to 41_{1v} is input to, for example, the distance measuring processing unit **101**.

Taking the pixels $10_{11}, 10_{21}$, and 10_{31} as an example, the read line of the pixel 10_{11} is connected to the first input port of the OR circuit 41_{11} , and the output of the OR circuit 41_{21} is connected to the second input port. The read line of pixel 10_{21} is connected to the first input port of the OR circuit 41_{21} , and the output of the OR circuit 41_{31} is connected to the second input port. The same applies to the OR circuit 41_{31} and subsequent circuits.

For the configuration illustrated in FIG. 9, for example, the vertical controller **102b** performs control so that simultaneous reading from respective pixels **10** in which positions correspond in the respective groups $12_u, 12_{u+1}, 12_{u+2}, \dots$ is not performed by using the signal EN_SPAD_V. In other words, the vertical controller **102b** controls so as to perform

12

reading from only one pixel **10** of a plurality of pixels **10** disposed every (v-1) pixels in a row. In the example of FIG. 9, the vertical controller **102b** performs control so that, for example, simultaneous reading from the pixel 10_{11} , the pixel 10_{21} , and the pixel 10_{31} is not performed. Not limited to this, the control of simultaneous reading in the column direction can also be performed by the horizontal controller **102a** using the signal EN_F.

On the other hand, in the configuration illustrated in FIG. 9, the vertical controller **102b** can designate simultaneous reading from v pixels **10** disposed consecutively in the column. At this time, the vertical controller **102b** can designate the pixels **10** from which reading is to be performed at the same time across the groups $12_u, 12_{u+1}, 12_{u+2}, \dots$. That is, in the configuration illustrated in FIG. 9, v consecutive pixels **10** in the column direction can be read at the same time. For example, it is possible to designate simultaneous reading from v pixels **10** disposed consecutively from the third pixel 10_{13} from the beginning included in the group 12_u to the second pixel 10_{22} from the beginning included in the group 12_{u+1} .

Further, when the vertical controller **102b** designates simultaneous reading from v pixels **10** disposed continuously in a column, the vertical controller **102b** controls so as not to perform reading from the other pixels **10** in the column. Therefore, for example, the output of the OR circuit 41_{11} is the signal Vpls read from any one pixel **10** of the pixels $10_{11}, 10_{21}, 10_{31}, \dots$.

In this way, by connecting the read line of each pixel **10** and performing read control on each pixel **10**, it is possible to reduce the number of read lines on a column basis.

Example of Pixel Array Scanning Method Using Existing Technology

Next, prior to the description of the present disclosure, a method of scanning the pixel array unit **100** by the existing technique will be schematically described. FIG. 10 is a diagram for explaining a method of scanning the pixel array unit **100** by the existing technique.

In the example of FIG. 10, in the pixel array unit **100**, each block $51a_1, 51a_2, 51a_3$, and $51a_4$ includes a total of (i×j) pixels **10** with i pixels in the column direction (height i) and j pixels in the row direction (width j). Scanning is performed on a scan region **50a** including the blocks $51a_1, 51a_2, 51a_3$, and $51a_4$ aligned in the row direction of the pixel array unit **100**.

With reference to FIG. 4, the distance measuring processing unit **101** converts by the conversion unit **110**, for example, each signal Vpls output from the pixels **10** included in the block $51a_1$ at each time of the light reception timing. The generation unit **111** adds the time information obtained by converting the signal Vpls by the conversion unit **110** in each predetermined time range to generate a histogram. That is, the block $51a_1$ represents an addition unit for adding time information when generating a histogram. Based on this histogram, the distance information at the representative position of the block $51a_1$ (for example, the center position of the block $51a_1$) can be acquired.

For example, the overall controller **103** performs the process for the block $51a_1$ on the blocks $51a_1, \dots$, and $51a_4$ aligned in the row direction of the target region in the pixel array unit **100**. The target region is, for example, a region in which enabled pixels **10** are disposed in the pixel array unit **100**. In this way, scanning is performed on the scan region where the height in the column direction matches the height of the block $51a_1$ in the column direction and the width in

13

the row direction matches the width of the target region. As a result, the distance information at the representative position of each block $51a_1, \dots, 51a_4$ can be obtained. This scanning is repeatedly performed while changing the position of the scan region $50a$ in the column direction of the target region on a scan region basis, and the distance information at each representative position in the target region in the pixel array unit 100 is acquired.

When the SPAD is used for the light receiving element 1000 of the pixel 10 , the SPAD requires a relatively high voltage for driving. Therefore, from the viewpoint of power consumption and circuit scale, the size of the scan region $50a$ for which reading is performed at one time has an upper limit. In addition, when the scanning in one scan region $50a$ is completed, and the next scanning is performed after the scan region $50a$ is moved in the column direction, the scan may be affected by the heat or the like of each light receiving element 1000 in the scan region where the scanning was performed immediately before. In addition, faster scanning for the entire target region may be required.

First Embodiment

Next, the first embodiment will be described. In the first embodiment, of the pixels 10 included in the target region of the pixel array unit 100 , the target pixels, each of which is a pixel 10 from which the pixel signal is read, are thinned out and designated with the pixel 10 as a unit. The target pixels thinned out and designated are scanned for each block in the scan region to obtain distance information.

FIG. 11 is a diagram for explaining a method of scanning the pixel array unit 100 according to the first embodiment. In FIG. 11 , a scan region $50b$ includes four blocks $51b_1, 51b_2, 51b_3$, and $51b_4$. In this example, each block $51b_1$ to $51b_4$ includes $(i \times 2)$ rows of the array of respective pixels 10 in the pixel array unit 100 , and has a height $(i \times 2)$ in the column direction. Each block $51b_1$ to $51b_4$ has a width j corresponding to j pixels 10 in the row direction, as in FIG. 10 described above.

Here, each block $51b_1$ to $51b_4$ is designated by thinning out the target pixels from which the pixel signal is read by one row. More specifically, each block $51b_1$ to $51b_4$ includes eight pixels 10 in the column direction (height $(i \times 2) = 4 \times 2$), and respective pixels 10 disposed in the c -th row, the $(c+2)$ th row, the $(c+4)$ th row, and the $(c+6)$ th row are designated as the target pixel from which reading is performed. On the other hand, respective pixels 10 disposed in the $(c+1)$ th row, the $(c+3)$ th row, the $(c+5)$ th row and the $(c+7)$ th row are not designated as the target pixel.

With reference to FIGS. 4 and 7 , for example, the overall controller 103 generates the signal EN_SPAD_V that designates on (read)/off (non-read) of respective rows included in the scan region $50b$ every other row, and generates the signal EN_SPAD_H that designates respective columns to turn on. The overall controller 103 passes the generated signals EN_SPAD_V and EN_SPAD_H to the vertical controller $102b$ and the horizontal controller $102a$, respectively.

Based on the passed signal EN_SPAD_V, the vertical controller $102b$ generates the signal XEN_SPAD_V, for each row, that turns on and off the rows every other row to supply the signal XEN_SPAD_V to the pixel array unit 100 . Based on the passed signal EN_SPAD_H, the horizontal controller $102a$ generates the signal XEN_SPAD_H, for each column, that turns on respective columns included in the target region to supply the signal XEN_SPAD_H to the pixel array unit 100 .

14

FIG. 12 is a diagram illustrating a comparison between the target pixel designated by the scanning method according to the first embodiment and the target pixel designated by the scanning method according to the existing technique described with reference to FIG. 10 . On the left side of FIG. 12 , an example of the target pixel designated by the scanning method according to the existing technique is illustrated. Further, on the right side of FIG. 12 , an example of the target pixel designated by the scanning method according to the first embodiment is illustrated.

Here, in the figure related to the existing technology on the left side of FIG. 12 , the scan region $50a$ includes four rows (height $i=4$). That is, each of the blocks $51a_1$ to $51a_4$ included in the scan region $50a$ has a predetermined width j in the row direction and includes four rows in the column direction. Therefore, each block $51a_1$ to $51a_4$ includes $(i \times j)$ target pixels.

On the other hand, in the figure according to the first embodiment on the right side of FIG. 12 , the scan region $50b$ includes 8 rows (height $(i \times 2)$). That is, each block $51b_1$ to $51b_4$ included in the scan region $50b$ has a predetermined width w in the row direction and includes eight rows in the column direction. Here, thinning is performed on each block $51b_1$ to $51b_4$ every other row. Therefore, each of the blocks $51b_1$ to $51b_4$ includes $(i \times j)$ target pixels, which are the same as those of each of the blocks $51a_1$ to $51a_4$ by the scanning method according to the existing technology.

In this way, the number of target pixels included in the addition unit that generates the histogram is equalized between the case where thinning is performed and the case where thinning is not performed. As a result, for example, the process in each unit (conversion unit 110 , generation unit 111 , and signal processing unit 112) of the distance measuring processing unit 101 can be standardized between the case where the thinning is performed and the case where the thinning is not performed.

In the example of the scanning method according to the first embodiment illustrated on the right side of FIGS. 11 and 12 , the target pixels are thinned out every other row and designated. For this reason, for example, the block $51b_1$ according to the first embodiment has the same number of the target pixels included as, for example, the block $51a_1$ in the case where thinning is not performed, while having a doubled height (number of rows) in the column direction on which scanning is performed.

Therefore, by performing thinning by the scanning method according to the first embodiment, the target region in the pixel array unit 100 can be performed at a higher speed than the target region when thinning is not performed. Further, by performing the thinning, it is possible to acquire distance information of a range wider than that when the thinning is not performed or acquire distance information of the same range with power consumption less than that when the thinning is not performed.

The distance measuring device 1 according to the first embodiment can switch between a first mode in which scanning by the target pixel designated after performing thinning is performed, and a second mode in which scanning by the target pixel designated without performing thinning is performed.

For example, the overall controller 103 sets, in response to an instruction, from the outside, as to whether to perform the first mode or the second mode, appropriately signals EN_SPAD_H, EN_SPAD_V, EN_PR, and the like according to the instructed mode of the first and second modes and passes them to the horizontal controller $102a$ and the vertical controller $102b$. The horizontal controller $102a$ and the

15

vertical controller **102b** designate the scan region and the target pixel in the scan region according to the signals EN_SPAD_H, EN_SPAD_V, EN_PR, and the like.

Here, when thinning is performed, the number of target pixels per area is reduced, the interval between the representative positions where the distance information is acquired is increased, and the resolution of the distance information is lowered, as compared with the case where the thinning is not performed. Therefore, there is a trade-off relationship between the scanning speed and power consumption, and the resolution when thinning is performed. Therefore, it is preferable to appropriately switch between the first mode and the second mode according to the purpose of distance measurement and the like.

The number of target pixels in each block when thinning is performed and the number of target pixels in each block when thinning is not performed do not necessarily have to be the same number. In this case, in order to maintain the consistency of the distance measurement results in distance measurement when thinning is performed and distance measurement when thinning is not performed, it is preferable that the number of target pixels of each block when thinning is performed and the number of target pixels of each block when thinning is not performed have substantially the same number. For example, the number of target pixels of each block when thinning is performed is such that the distance measurement result of distance measurement when thinning is performed is within an allowable error range with respect to the distance measurement result when thinning is not performed.

Mitigation of Influence From Adjacent Blocks by Thinning According to First Embodiment

By thinning out the target pixels according to the first embodiment, it is possible to mitigate the influence of heat and the like received from adjacent blocks. FIG. 13 is a diagram for explaining mitigation of the influence from the adjacent block by thinning out the target pixels according to the first embodiment.

Consideration is made in which after scanning the scan region **50b** illustrated in FIG. 13, scanning is performed on the scan region including the block **51b₄'** which is adjacent, in the column direction, to the block **51b₄** included in the scan region **50b**. In the blocks **51b₄** and **51b₄'**, respective pixels **10** disposed in the first row, the third row, the fifth row, and the seventh row from the lower end are designated as target pixels. On the other hand, respective pixels **10** disposed in the second row, the fourth row, the sixth row, and the eighth row from the lower end are not designated as target pixels. It is assumed that the supply of the power supply voltage Vdd to the light receiving element **1000** is stopped for the respective pixels **10** that are not designated as the target pixels, for example, by controlling by the signal EN_PR.

In this case, respective pixels **10** disposed at the first row, which is the lower end row of the block **51b₄'** are designated as the target pixels. On the other hand, in the eighth row, which is the upper end row of the block **51b₄**, included in the scan region **50b** where the operation was completed immediately before, which is adjacent to the first row, respective pixels **10** disposed are not designated as the target pixels, and the supply of the power supply voltage Vdd is stopped.

In this way, by performing thinning on a row basis, for example, a row in which respective pixels **10** that are not designated as target pixels and to which the supply of the power supply voltage Vdd is stopped are disposed can be

16

disposed at the boundary between block **51b₄'**, which is currently being scanned, and block **51b₄**, which was scanned immediately before. Therefore, it is possible to mitigate the influence of the heat generated by respective pixels **10** of the block **51b₄** due to the immediately preceding scan on the block **51b₄'**, which is to be scanned.

Reading of Signal Vpls From Respective Pixels **10** According to First Embodiment

Next, the reading process of the signal Vpls from respective pixels **10** according to the first embodiment will be described. In the first embodiment, as described with reference to FIG. 9, the read line that reads the signal Vpls is shared every v pixels **10** disposed along the row, and $(v-1)$ pixels **10** disposed consecutively in the column direction can be read at the same time. The thinning corresponding to the sharing of the read line will be described.

FIGS. 14A and 14B are diagrams corresponding to FIG. 9 described above, and are diagrams for explaining thinning corresponding to the sharing of the read line according to the first embodiment. In addition, in FIGS. 14A and 14B, the black square indicates pixels **10** controlled by the vertical controller **102b** so that reading from the pixels **10** is not performed, or the pixels **10** are turned off, and the white square indicate pixels **10** that are controlled so that reading from the pixels **10** is performed.

FIG. 14A illustrates an example in which slipping is performed every other row, for example, as in the above-mentioned blocks **51b₁** to **51b₄**. Here, it is assumed that $v=7$, the read lines are shared every six $(=v-1)$ pixels **10**, and seven pixels **10** can be read simultaneously. Further, it is assumed that, for example, the block **51b₁** that is thinned out every other row includes eight pixels **10** in the column direction (height $(i \times 2)=8$). In this case, the pixels **10** in a row included in block **51b₁** correspond to, for example, the pixels **10₁₁** to **10_{1v}** and the pixel **10₂₁** in FIG. 14A, and are disposed across the group **12_u** and the group **12_{u+1}**.

Here, in the example of FIG. 14A, the group **12_u** includes an odd number of pixels **10₁₁** to **10_{1v}**. For this reason, in performing thinning every other row, when the initial pixel **10₁₁** of the group **12_u** is a target pixel to which reading is designated with the right side of the figure as the beginning of the position in FIG. 14A, target pixels and non-target pixels are disposed alternately such that a pixel **10** adjacent to the left is a non-target pixel to which reading is not designated, a pixel **10** adjacent to the above left is a target pixel, . . . , and the rear end pixel **10_{1v}** is the target pixel. On the other hand, in the group **12_{u+1}** adjacent to the group **12_u**, the order of the target pixels and the non-target pixels is reversed relative to the group **12_u**.

In the case of this example in which thinning is performed every other row, the target pixel is designated with two pixels in the column direction as a cycle. Therefore, in 14 pixels **10** with a cycle of seven pixels $\times 2$ pixels disposed consecutively in the column direction, that is, in two adjacent groups **12_u** and **12_{u+1}**, reading is performed from any one pixel of a set of pixels **10** in which the read line is shared (for example, a set of pixel **10₁₁** and pixel **10₂₁**, a set of pixel **10_{1v}** and pixel **10_{2v}**, . . .). This also applies to the other pixels **10** included in the group **12_u**, the group **12_{u+1}**,

By designating the rows to be skipped in this way and the pixels **10** from which reading can be performed at the same time, it is possible to perform reading by thinning while sharing the read line.

In the above, an example where thinning is performed every other row is described, but the embodiment is not

17

limited to this example, and it is possible to skip more rows. FIG. 14B illustrates an example in which two rows at a time are skipped for block (for example, block 51a1) on which thinning is not performed. In this case, the target pixel is designated with three pixels in the column direction as a cycle. Therefore, in 21 pixels 10 with a cycle of seven pixels×three pixels disposed continuously in the column direction, that is, in three adjacent groups 12_u, 12_{u+1}, and 12_{u+2}, reading is performed from any one pixel of a set of pixels 10 in which the read line is shared (for example, a set of pixels 10₁₁, 10₂₁, and 10₃₁, a set of pixels 10_{1v}, 10_{2v}, and 10_{3v}, . . .). This also applies to the other pixels 10 included in the group 12_u, the group 12_{u+1}, the group 12_{u+2}, . . .

In the above description, no thinning, thinning of every other row, and thinning of two rows at a time have been described, but the interval of thinning is not limited to these examples. That is, the thinning interval may be another value under the condition that the number of rows to be read after thinned out does not match the interval of the pixels 10 sharing the read line. For example, when the read line is shared at (v-1) intervals in the column direction, the thinning interval can be the interval excluding the (v-1) rows.

Here, for respective pixels 10 disposed continuously in the column, the number of pixels capable of simultaneous reading by sharing the read line is defined as the number of pixels (w+1). Further, a value g when the next reading after reading a row (pixel 10) is performed on the g-th row with the row as a base point is defined as the number of thinning g. For example, when each row is continuously read without performing thinning, the number of thinning g=1. In addition, when thinning is performed every other row as illustrated in FIG. 14A, the number of thinning g=2.

When the number of pixels (w+1) that can be read simultaneously and the number of thinning g are defined in this way, the number of pixels in the maximum read region S_{MAX} where simultaneous reading can be performed is calculated by the following equation (2).

$$S_{MAX}=g \times (w+1) \quad (2)$$

That is, when the number of pixels from which simultaneous reading can be performed is (w+1)=7, and the number of thinning g is one (without thinning), the number of pixels in the maximum read region S_{MAX}=1×7=7 (pixels). When the number of thinning g is two (perform thinning every other row), the number of pixels in the maximum read region S_{MAX}=2×7=14 (pixels).

Further, the number of the row read lines R_{CLM} that supplies the signal V_{pls} read from each pixel 10 to each read line in the column direction is obtained by the following equation (3).

$$R_{CLM}=(g \times w)+1 \quad (3)$$

That is, when the number of pixels from which simultaneous reading can be performed is (w+1)=7, and the number of thinning g=one (without performing thinning), the number of row read lines R_{CLM}=(1×6)+1=7 (lines). When the number of thinning is g=two (performing thinning every other row), the number of row read lines R_{CLM}=(2×6)+1=13 (lines). The number of read lines R_{CLM} indicates that non-target pixels are disposed in a predetermined number of rows at the upper end or the lower end of the block when thinning is performed.

First Modification of First Embodiment

Next, a first modification of the first embodiment will be described. In the above, thinning is performed on a row

18

basis, that is, in the column direction of the target region in the pixel array unit 100, but the embodiment is not limited to this example. FIG. 15 is a diagram schematically illustrating an example of thinning according to the first modification of the first embodiment.

The first modification of the first embodiment is an example in which columns are skipped in the row direction as illustrated in FIG. 15. More specifically in the example of FIG. 15, thinning is performed every other column for, for example, each of the blocks 51a₁ to 51a₄ with a height i and a width j illustrated in FIG. 10 to generate blocks 51c₁, 51c₂, 51c₃, and 51c₄ each having a height (i×2) and a width j corresponding to the blocks 51a₁ to 51a₄, respectively. A scan region 50c includes these blocks 51c₁ to 51c₄.

This will be described more specifically. As an example, it is assumed that each block 51c₁ to 51c₄ includes 9 pixels 10 in the row direction (width j=9). The respective pixels 10 disposed in the r-th column, the (r+2)th column, the (r+4)th column, the (r+6)th row, and the (r+8)th row are designated as the target pixels, and the respective pixels 10 disposed in the (r+1)th column, the (r+3)th column, the (r+5)th column, and the (r+7)th column are not designated as the target pixels.

With reference to FIGS. 4 and 7, for example, the overall controller 103 generates the signal EN_SPAD_V that designates each row included in the scan region 50c to turn on, and generates the signal EN_SPAD_H that designates each column to turn on (read)/off (non-read) every other column. The overall controller 103 passes the generated signals EN_SPAD_V and EN_SPAD_H to the vertical controller 102b and the horizontal controller 102a, respectively.

The vertical controller 102b generates the signal XEN_SPAD_V, for each row, that turns on respective rows included in the target region based on the passed signal EN_SPAD_V to supply the signal XEN_SPAD_V to the pixel array unit 100. Based on the passed signal EN_SPAD_H, the horizontal controller 102a generates the signal XEN_SPAD_H, for each column, that turns on and off respective columns included in the target region on a column basis to supply the signal XEN_SPAD_H to the pixel array unit 100.

In the case of this example, it is conceivable that the signal EN_PR that controls the supply of the power supply voltage V_{dd} to each light receiving element 1000 is generated by the horizontal controller 102a and supplied to each row.

In the case of the first modification of the first embodiment, considering the sharing of the read line described with reference to FIG. 9, it is conceivable to use a method in which, for example, pixels are divided in the column direction on a basis of the number of pixels 10 (the number of rows) from which simultaneous reading can be performed to perform scanning.

Here, in the example of FIG. 15, since the number of columns of each block 51c₁ to 51c₄ is odd, the number of pixels 10 designated as the target pixels in each block 51c₁ to 51c₄ is larger than, for example, the number of pixels 10 included as the target pixels in each block 51a₁ to 51a₄ in FIG. 10 by i pixels. Whether to allow the difference in the number of pixels 10, designated as the target pixels, which occurs between when thinning is not performed and when thinning is performed, depends on the specification of the system and the like.

Second Modification of First Embodiment

Next, a second modification of the first embodiment will be described. In the above, thinning is performed on a row

basis or on a column basis, but the embodiment is not limited to this example. FIG. 16 is a diagram schematically illustrating an example of thinning according to the second modification of the first embodiment.

The second modification of the first embodiment is an example in which thinning is performed in each of the row direction and the column direction as illustrated in FIG. 16. More specifically, thinning is performed every other pixel 10 in the row direction and in the column direction on, for example, each block 51a₁ to 51a₄ with a height i and a width j illustrated in FIG. 10 to generate blocks 51d₁, 51d₂, 51d₃, and 51d₄ each having a height (ix2) and a width j corresponding to the blocks 51a₁ to 51a₄, respectively. A scan region 50d includes these blocks 51d₁ to 51d₄.

In the example of FIG. 16, in each of the blocks 51d₁ to 51d₄, the target pixels indicated by diagonal lines and the non-target pixels indicated by white are disposed in a checkered pattern. According to this thinning method, the target pixels are disposed so as not to be adjacent to each other in the column direction and the row direction.

This will be described more specifically. As an example, it is assumed that the block 51d₁ includes 9 pixels 10 in the row direction (width j=9). In the r-th column, the (r+2)th column, the (r+4)th column, the (r+6)th row and the (r+8)th row, the (c+1)th row, the (c+3)th row, the (c+5)th row, and the (c+7)th row are designated as the target pixels. Also, in the (r+1)th column, the (r+3)th column, the (r+5)th column, and the (r+7)th row, the c-th row, the (c+2)th row, the (c+4)th row, the (c+6)th row, and the (c+8)th row are designated as the target pixels.

The block 51d₃ and the block 51d₁ are thinned in the same manner, and the blocks 51d₂ and 51d₄ are thinned in a manner in which the positions of the designated target pixels are obtained by inverting corresponding positions in the blocks 51d₁ and 51d₃.

In the second modification of the first embodiment, for example, it is conceivable to perform control so that scanning is performed on the same scan region 50d twice with the designated target pixels differentiated. With reference to FIGS. 4 and 7, for example, the overall controller 103 generates the signal EN_SPAD_V that designates each row included in the scan region 50d to turn on every other row, and generates the signal EN_SPAD_H that designates each column included in the scan region 50d to turn on every other column.

More specifically, for example, the overall controller 103 generates the signal EN_SPAD_V that designates the (c+1)th row, the (c+3)th row, the (c+5)th row, and the (c+7)th row to turn on. In addition, the overall controller 103 generates the signal EN_SPAD_H that designates the r-th column, the (r+2)th column, the (r+4)th column, the (r+6)th row, and the (r+8)th row of each block 51d₁ to 51d₄ to turn on. The overall controller 103 passes the generated signals EN_SPAD_V and EN_SPAD_H to the vertical controller 102b and the horizontal controller 102a, respectively.

Based on the passed signal EN_SPAD_V, the vertical controller 102b supplies, to the pixel array unit 100 the signal XEN_SPAD_V that turns on the (c+1)th row, the (c+3)th row, the (c+5)th row and the (c+7)th row, and turns off the c-th row, the (c+2)th row, the (c+4)th row, the (c+6)th row, and the (c+8)th row of respective rows included in the target region. Further, based on the passed signal EN_SPAD_H, the horizontal controller 102a supplies, to the pixel array unit 100, the signal XEN_SPAD_H that turns on the r-th column, the (r+2)th column, the (r+4)th column, the (r+6)th row and the (r+8)th row, and turns off the (r+1)th

column, the (r+3)th column, the (r+5)th column, the (r+7)th row, and the (r+8)th row of each column included in the target region.

Further, the vertical controller 102b supplies the signal EN_PR that controls the supply of the power supply voltage Vdd to each light receiving element 1000 to the c-th row, the (c+2)th row, the (c+4)th row, the (c+6)th row, and the (c+8)th row.

As described above, the overall controller 103 performs control so that scanning is performed on the scan region 50d on which thinning is performed every other row and thinning is performed every other column. After this scan, the overall controller 103 inverts the designated positions of the target pixels in the scan region 50d, that is, the overall controller 103 sets the target pixel in a previous scan to the non-target pixel, and the asymmetrical pixel in the previous scan to the target pixel in the scan region 50d, and performs an operation for the scan region 50d again.

Also, in the second modification of this first embodiment, as in the first modification described above, the number of pixels 10 designated as the target pixels in each block 51d₁ to 51d₄ may differ from the number of pixels 10 included as the target pixels in each of the blocks 51a₁ to 51a₄ in FIG. 10, for example. Whether to allow the difference in the number of pixels 10, designated as the target pixels, which occurs between when thinning is not performed and when thinning is performed, depends on the specification of the system and the like.

Second Embodiment

Next, the second embodiment will be described. In the first embodiment described above and each modification thereof, the target pixel to be scanned is designated with the pixel 10 as a unit. In the second embodiment, the target pixel is designated for each block, and thinning is performed between the blocks.

FIG. 17 is a diagram for explaining a method of scanning the pixel array unit 100 according to the second embodiment. In FIG. 17, a scan region 50e₁ includes four blocks 51e₁₁, 51e₁₂, 51e₁₃, and 51e₁₄. In this example, each block 51e₁₁ to 51e₁₄ includes v rows of the array of respective pixels 10 in the pixel array unit 100, and has a height v in the column direction. It is assumed that each block 51e₁₁ to 51e₁₄ has a width j corresponding to the j pixels 10 in the row direction as in FIG. 10 described above. Further, all the pixels 10 included in the blocks 51e₁₁ to 51e₁₄ are designated as the target pixels.

After scanning the scan region 50e₁, the overall controller 103 skips a predetermined number of rows in the column direction and scans the next scan region 50e₂. As in the scan region 50e₁, the scan region 50e₂ includes four blocks 51e₂₁/51e₂₂, 51e₂₃, and 51e₂₄ each having a height v and a width j. As described above, in the example of FIG. 17, the overall controller 103 scans the scan region 50e₁ and then skips a predetermined number of rows to scan the next scan region 50e₂. The number of rows to be skipped is not particularly limited. In the example of FIG. 17, thinning is performed by the same number of rows (v rows) as those of the scan regions 50e₁ and 50e₂.

FIG. 18 is a diagram corresponding to FIG. 9 described above, and is a diagram for explaining thinning corresponding to the sharing of the read line according to the second embodiment. In addition, in FIGS. 14A and 14B, the black square indicates pixels 10 controlled by the vertical controller 102b so that reading from the pixels 10 is not performed,

21

or the pixels **10** are turned off, and the white square indicate pixels **10** that are controlled so that reading from the pixels **10** is performed.

As described above, the read line for reading the signal V_{pl} is shared for v pixels **10** in the column direction, and simultaneous reading from the v pixels **10** continuously disposed in the column direction can be performed. In FIG. **18**, it is illustrated that the v pixels **10**₁₁ to **10**_{1v} are designated for reading at the same time. Therefore, it is preferable that the height of the scan regions **50e**₁, **50e**₂, . . . be a height corresponding to the v pixels **10** or less than the v pixels **10**, which allows simultaneous reading.

According to this second embodiment, the number of target pixels included in each block **51e**₁₁ to **51e**₁₄, . . . is equal to the number of target pixels included in the block (for example, block **51a**₁) when thinning is not performed. Therefore, the accuracy of distance measurement in each block **51e**₁₁ to **51e**₁₄, . . . is the same as that of a block when thinning is not performed. Since the thinning is performed on a block basis in the scanning method according to the second embodiment, the scanning method according to the second embodiment is disadvantageous in terms of resolution as compared with that according to the first embodiment described above.

Modification of Second Embodiment

Next, a modification of the second embodiment will be described. FIG. **19** is a diagram schematically illustrating an example of thinning according to the modification of the second embodiment. As illustrated in FIG. **19**, the modification of the second embodiment is an example in which the target pixels are designated on a block basis, and when performing thinning between blocks, the target region of the pixel array unit **100** is thinned out in the row direction and the column direction on a block basis.

In FIG. **19**, for example, a scan region **50f**₁ includes four blocks **51f**₁, **51f**₂, **51f**₃, and **51f**₄ each having a height v and a width j . In blocks **51f**₁ and **51f**₃ of these, all the included pixels **10** are designated as target pixels. On the other hand, in the blocks **51f**₂ and **51f**₄, all the included pixels **10** are non-target pixels.

In the scan region **50f**₂ immediately above the scan region **50f**₁, all of the pixels **10** included in the blocks corresponding to the blocks **51f**₁ and **51f**₃ in the scan region **50f**₁ are the non-target pixels, and all of the pixels **10** included in the blocks corresponding to the blocks **51f**₂ and **51f**₄ are the target pixels. Further, the scan region **50f**₃ immediately above the scan region **50f**₂ has the same configuration as the scan region **50f**₁, and the scan region **50f**₄ immediately above the scan region **50f**₃ has the same configuration as the scan region **50f**₂.

Also, in the modification of this second embodiment, the number of target pixels included in each block **51f**₁ to **51f**₄, . . . is equal to the number of target pixels included in the block (for example, block **51a**₁) when thinning is not performed. Therefore, the accuracy of distance measurement in each block **51f**₁ to **51f**₄, . . . is the same as that of a block when thinning is not performed. Since the thinning is performed on a block basis in the scanning method according to the modification of the second embodiment, the scanning method according to the second embodiment is disadvantageous in terms of resolution as compared with that according to the first embodiment described above.

Third Embodiment

Next, as the third embodiment of the present disclosure, the first embodiment of the present disclosure and each

22

modification thereof, and the application example of the modification of the second embodiment will be described. FIG. **20** is a diagram illustrating a usage example in which the distance measuring device **1** according to each modification of the first embodiment and the first embodiment described above according to the second embodiment is used.

The distance measuring device **1** described above can be used in various cases in which light such as visible light, infrared light, ultraviolet light, and X-ray is sensed as described below.

A device that captures images used for appreciation, such as a digital camera and a mobile device with a camera function.

A device used for traffic, such as an in-vehicle sensor that images the front, rear, surroundings, and interior of an automobile, a surveillance camera that monitors traveling vehicles and roads, a distance measuring sensor that measures a distance between vehicles, and the like, for safe driving such as automatic stop and recognition of the driver's condition, etc.

A device used for home appliances, such as a TV, a refrigerator, and an air conditioner, to take a picture of a user's gesture and operate the device according to the gesture.

A device used for medical treatment and healthcare, such as an endoscope and a device that performs angiography by receiving infrared light.

A device used for security, such as a surveillance camera for crime prevention and a camera for personal authentication.

A device used for beauty, such as a skin measuring device that photographs the skin and a microscope that photographs the scalp.

A device used for sports, such as an action camera and a wearable camera for sports applications.

A device used for agriculture, such as a camera for monitoring the condition of fields and crops.

Further Application Example of Technology
According to Present Disclosure

Example of Application to Moving Object

The technology according to the present disclosure may be further applied to devices mounted on various moving objects such as automobiles, electric vehicles, hybrid electric vehicles, motorcycles, bicycles, personal mobility, airplanes, drones, ships, and robots.

FIG. **21** is a block diagram illustrating a schematic configuration example of a vehicle control system, which is an example of a moving object control system to which the technique according to the present disclosure can be applied.

A vehicle control system **12000** includes a plurality of electronic control units connected via a communication network **12001**. In the example illustrated in FIG. **21**, the vehicle control system **12000** includes a drive system control unit **12010**, a body system control unit **12020**, a vehicle-exterior information detection unit **12030**, an in-vehicle information detection unit **12040**, and an integrated control unit **12050**. Further, as the functional configuration of the integrated control unit **12050**, a microcomputer **12051**, an audio image output unit **12052**, and an in-vehicle network I/F (interface) **12053** are illustrated.

The drive system control unit **12010** controls the operation of the device related to the drive system of the vehicle according to various programs. For example, the drive

system control unit **12010** serves as a driving force generation unit that generates the driving force of the vehicle such as an internal combustion engine or a driving motor, a driving force transmission mechanism that transmits the driving force to the wheels, a steering mechanism for adjusting a steering angle of the vehicle, and a control device such as a braking device that generates a braking force of the vehicle.

The body system control unit **12020** controls the operation of various devices mounted on the vehicle body according to various programs. For example, the body system control unit **12020** functions as a control device for a keyless entry system, a smart key system, a power window device, or various lamps such as a head lamp, a back lamp, a brake lamp, a blinker and a fog lamp. In this case, the body system control unit **12020** may receive radio waves transmitted from a portable device that substitutes for the key or signals of various switches. The body system control unit **12020** receives the input of these radio waves or signals and controls a door lock device, a power window device, a lamp, and the like of the vehicle.

The vehicle-exterior information detection unit **12030** detects information outside the vehicle equipped with the vehicle control system **12000**. For example, an imaging unit **12031** is connected to the vehicle-exterior information detection unit **12030**. The vehicle-exterior information detection unit **12030** causes the imaging unit **12031** to capture an image of the outside of the vehicle and receives the picked up image. The vehicle-exterior information detection unit **12030** may perform the object detection process or the distance detection process of detecting a person, a vehicle, an obstacle, a sign, or characters on the road surface based on the received image. The vehicle-exterior information detection unit **12030** performs the image process on the received image, for example, and performs the object detection process and the distance detection process based on the result of the image process.

The imaging unit **12031** is an optical sensor that receives light to output an electrical signal according to the amount of the light received. The imaging unit **12031** can output an electrical signal as an image or can output it as distance measurement information. Further, the light received by the imaging unit **12031** may be visible light or invisible light such as infrared light.

The in-vehicle information detection unit **12040** detects in-vehicle information. For example, a driver state detector **12041** that detects the driver's state is connected to the in-vehicle information detection unit **12040**. The driver state detector **12041** includes, for example, a camera that captures the driver, and the in-vehicle information detection unit **12040** may calculate the degree of fatigue or concentration of the driver, or may determine whether the driver is dozing based on the detection information input from the driver state detector **12041**.

The microcomputer **12051** can calculate the control target value of the driving force generation unit, the steering mechanism or the braking device based on the information inside and outside the vehicle acquired by the vehicle-exterior information detection unit **12030** or the in-vehicle information detection unit **12040** to output a control command to the drive system control unit **12010**. For example, the microcomputer **12051** can perform cooperative control for the purpose of realizing a function of an advanced driver assistance system (ADAS) including vehicle collision avoidance or impact mitigation, follow-up driving based on inter-vehicle distance, vehicle speed maintenance driving, vehicle collision warning, or vehicle lane deviation warning.

In addition, based on the information around the vehicle acquired by the vehicle-exterior information detection unit **12030** or the in-vehicle information detection unit **12040**, the microcomputer **12051** can perform cooperative control for the purpose of automatic driving or the like in which the vehicle travels autonomously without depending on the operation of the driver by controlling the driving force generation unit, the steering mechanism, the braking device, etc.

Further, the microcomputer **12051** can output a control command to the body system control unit **12020** based on the information outside the vehicle acquired by the vehicle-exterior information detection unit **12030**. For example, the microcomputer **12051** can control the head lamps according to the position of the preceding vehicle or the oncoming vehicle detected by the vehicle-exterior information detection unit **12030** to perform cooperative control for the purpose of anti-glare such as switching the high beam to the low beam.

The audio image output unit **12052** transmits an output signal of at least one of an audio and an image to an output device capable of visually or audibly notifying the passenger or the outside of the vehicle of information. In the example of FIG. **21**, an audio speaker **12061**, a display unit **12062**, and an instrument panel **12063** are exemplified as output devices. The display unit **12062** may include, for example, at least one of an onboard display and a heads-up display.

FIG. **22** is a diagram illustrating an example of the installation position of the imaging unit **12031**. In FIG. **22**, a vehicle **12100** has imaging units **12101**, **12102**, **12103**, **12104**, and **12105** as the imaging unit **12031**.

For example, the imaging units **12101**, **12102**, **12103**, **12104**, and **12105** are provided at positions such as the front nose, the side mirrors, the rear bumper, the back door, and the upper part of the windshield in the vehicle interior of the vehicle **12100**. The imaging unit **12101** provided on the front nose and the imaging unit **12105** provided on the upper part of the windshield in the vehicle interior mainly acquire an image in front of the vehicle **12100**. The imaging units **12102** and **12103** provided on the side mirrors mainly acquire images of the sides of the vehicle **12100**. The imaging unit **12104** provided on the rear bumper or the back door mainly acquires an image behind the vehicle **12100**. The front image acquired by the imaging units **12101** and **12105** is mainly used for detecting a preceding vehicle or a pedestrian, an obstacle, a traffic light, a traffic sign, a lane, or the like.

Note that FIG. **22** illustrates an example of the shooting range of the imaging units **12101** to **12104**. An imaging range **12111** indicates the imaging range of the imaging unit **12101** provided on the front nose, imaging ranges **12112** and **12113** indicate the imaging ranges of the imaging units **12102** and **12103** provided on the side mirrors, respectively, and an imaging range **12114** indicates the imaging range of the imaging unit **12104** provided on the rear bumper or the back door. For example, by superimposing the image data imaged by the imaging units **12101** to **12104**, a bird's-eye view image of the vehicle **12100** as viewed from above can be obtained.

At least one of the imaging units **12101** to **12104** may have a function of acquiring distance information. For example, at least one of the imaging units **12101** to **12104** may be a stereo camera composed of a plurality of imaging devices, or may be an imaging device having pixels for phase-difference detection.

For example, by finding the distance to each three-dimensional object within the imaging ranges **12111** to

25

12114, and the temporal change of this distance (relative velocity with respect to the vehicle 12100) based on the distance information obtained from the imaging units 12101 to 12104, the microcomputer 12051 can extract, in particular, a three-dimensional object that is the closest three-dimensional object on the traveling path of the vehicle 12100 and that travels in substantially the same direction as the vehicle 12100 at a predetermined speed (for example, 0 km/h or more) as a preceding vehicle. Further, the microcomputer 12051 can set an inter-vehicle distance to be secured in front of the preceding vehicle in advance, and can perform automatic braking control (including follow-up stop control), automatic acceleration control (including follow-up start control), and the like. In this way, cooperative control can be performed for the purpose of automatic driving or the like in which the vehicle travels autonomously without depending on the driver's operation.

For example, the microcomputer 12051 can sort three-dimensional object data related to a three-dimensional object into a two-wheeled vehicle, an ordinary vehicle, a large vehicle, a pedestrian, and other three-dimensional objects such as a utility pole based on the distance information obtained from the imaging units 12101 to 12104 to extract them, and can use them for automatic avoidance of obstacles. For example, the microcomputer 12051 identifies obstacles around the vehicle 12100 as an obstacle that are visible to the driver of the vehicle 12100 and an obstacle that are difficult to see. The microcomputer 12051 can determine the collision risk, which indicates the risk of collision with each obstacle, and when the collision risk is above the set value and there is a possibility of collision, the microcomputer 12051 can provide a driving assistance for collision avoidance by outputting an alarm to the driver via the audio speaker 12061 or the display unit 12062, or by performing forced deceleration and avoidance steering via the drive system control unit 12010.

At least one of the imaging units 12101 to 12104 may be an infrared camera that detects infrared rays. For example, the microcomputer 12051 can recognize a pedestrian by determining whether the pedestrian is present in the picked up images of the imaging units 12101 to 12104. Such pedestrian recognition includes, for example, a procedure for extracting feature points in picked up images of the imaging units 12101 to 12104 as an infrared camera, and a procedure of performing a pattern matching process on a series of feature points indicating the outline of an object to determine whether the object is a pedestrian. The microcomputer 12051 determines that a pedestrian is present in the picked up images of the imaging units 12101 to 12104, and when the pedestrian is recognized, the audio image output unit 12052 causes the display unit 12062 to superimpose and display a square outline for emphasis on the recognized pedestrian. Further, the audio image output unit 12052 may cause the display unit 12062 to display an icon or the like indicating the pedestrian at a desired position.

An example of the vehicle control system to which the technique according to the present disclosure can be applied is described above. The technique according to the present disclosure can be applied to, for example, the imaging unit 12031 of the configuration described above. Specifically, the distance measuring device 1 according to the first embodiment of the present disclosure described above can be applied to the imaging unit 12031. By applying the technique according to the present disclosure to the imaging unit 12031, it is possible to increase the distance measuring speed and save power of the distance measuring device 1 that measures the distance from a traveling vehicle.

26

Further, the effects in each embodiment described in the present specification are merely examples and are not limited, and other effects may be present.

Note that the present technology may also be configured as below.

(1) A measurement device comprising:

a light receiving unit including a plurality of pixels disposed in a target region in a matrix array;

a controller that designates one or more target pixels, of the plurality of pixels, from which a signal is to be read; and

a signal processing unit that performs a signal process based on the signal read from the one or more target pixels included in a block having a predetermined width in a row direction of the array and having a variable height in a column direction, wherein

the controller

switches, according to an instruction, between a first mode in which the one or more target pixels are designated after performing thinning on the plurality of pixels, and a second mode in which the one or more target pixels are designated without performing the thinning.

(2) The measurement device according to the above (1), wherein

the controller

designates the one or more target pixels with the block as a unit, and controls the thinning so that the number of the one or more target pixels included in the block in the first mode is substantially equal to the number of the one or more target pixels included in the block in the second mode.

(3) The measurement device according to the above (1) or (2), wherein

the controller

designates the one or more target pixels with the block as a unit.

(4) The measurement device according to any one of the above (1) to (3), wherein

the controller

performs, in the first mode, the thinning on a column direction of the array.

(5) The measurement device according to any one of the above (1) to (3), wherein

the controller

performs, in the first mode, the thinning on a row direction of the array.

(6) The measurement device according to any one of the above (1) to (3), wherein

the controller

performs, in the first mode, the thinning on a column direction and a row direction of the array.

(7) The measurement device according to any one of the above (1) to (6), wherein

each of a plurality of signal lines that transmits respective signals which are read from the plurality of pixels which is provided in the light receiving unit along a column direction of the array

is connected to two or more pixels, of the plurality of pixels, disposed along a column direction of the array every (v-1) pixels (where v is an integer of two or more) along the column direction of the array, and

the controller

designates the thinning at intervals at which the reading is not performed simultaneously from the two or more pixels.

27

(8) The measurement device according to any one of the above (1) to (7), wherein the controller stops power supply to a pixel, of the plurality of pixels, which is not designated as the one or more target pixels. 5

(9) The measurement device according to any one of the above (1) to (8), wherein the controller switches between the first mode and the second mode according to the instruction from an outside. 10

(10) A distance measuring device comprising:
 a light receiving unit including a plurality of pixels disposed in a target region in a matrix array;
 a controller that designates one or more target pixels, of the plurality of pixels, from which a signal is to be read; 15
 a time acquisition unit that measures a time from light emission timing when a light source emits light to light reception timing when the pixel included in a block having a predetermined width in a row direction of the array and having a variable height in a column direction receives the light to acquire a measured value; 20
 a generation unit that adds the number of the measured values in each predetermined time range based on the measured values to generate a histogram related to the block; and 25
 a calculation unit that calculates a distance to an object to be measured based on the histogram, wherein the controller switches, according to an instruction, between a first mode in which the one or more target pixels are designated after performing thinning on the plurality of pixels, and a second mode in which the one or more target pixels are designated without performing the thinning. 30

(11) The distance measuring device according to the above (10), wherein the controller designates the one or more target pixels with the block as a unit, and controls the thinning so that the number of the one or more target pixels included in the block in the first mode is substantially equal to the number of the one or more target pixels included in the block in the second mode. 40

(12) The distance measuring device according to the above (10) or (11), wherein the controller designates the one or more target pixels with the block as a unit. 45

(13) The distance measuring device according to any one of the above (10) to (12), wherein the controller performs, in the first mode, the thinning on a column direction of the array. 50

(14) The distance measuring device according to any one of the above (10) to (12), wherein the controller performs, in the first mode, the thinning on a row direction of the array. 55

(15) The distance measuring device according to any one of the above (10) to (12), wherein the controller performs, in the first mode, the thinning on a column direction and a row direction of the array. 60

(16) The distance measuring device according to any one of the above (10) to (15), wherein each of a plurality of signal lines that transmits respective signals which are read from the plurality of pixels 65

28

which is provided in the light receiving unit along a column direction of the array is connected to two or more pixels, of the plurality of pixels, disposed along a column direction of the array every (v-1) pixels (where v is an integer of two or more) along the column direction of the array, and the controller designates the thinning at intervals at which the reading is not performed simultaneously from the two or more pixels.

(17) The distance measuring device according to any one of the above (10) to (16), wherein the controller stops power supply to a pixel, of the plurality of pixels, which is not designated as the one or more target pixels.

(18) The distance measuring device according to any one of the above (10) to (17), wherein the controller switches between the first mode and the second mode according to the instruction from an outside.

REFERENCE SIGNS LIST

1 DISTANCE MEASURING DEVICE
2 LIGHT SOURCE UNIT
3 STORAGE UNIT
4 CONTROLLER
6 ELECTRONIC DEVICE
10, 10₁₁, 10₁₃, 10_{1v}, 10₂₁, 10₂₂, 10_{2v}, 10₃₁, 10_{3v} PIXEL
11 ELEMENT
41₁₁, 41_{1v}, 41₂₁, 41_{2v}, 41₃₁, 41₃₇ OR CIRCUIT
50a, 50b, 50c, 50d, 50e₁, 50e₂, 50f₁, 50f₂, 50f₃, 50f₄
 SCAN REGION
51a₁, 51a₂, 51a₃, 51a₄, 51b₁, 51b₂, 51b₃, 51b₄, 51b₄', 51c₁, 51c₂, 51c₃, 51c₄, 51d₁, 51d₂, 51d₃, 51d₄, 51e₁₁, 51e₁₂, 51e₁₃, 51e₁₄, 51e₂₁, 51e₂₂, 51e₂₃, 51e₂₄, 51f₁, 51f₂, 51f₃, 51f₄ BLOCK
100 PIXEL ARRAY UNIT
102 PIXEL CONTROLLER
102a HORIZONTAL CONTROLLER
102b VERTICAL CONTROLLER
103 OVERALL CONTROLLER

The invention claimed is:

1. A measurement device, comprising:
 a light receiving unit that includes a plurality of pixels in a target region in a matrix array, wherein the plurality of pixels includes a plurality of read lines;
 a controller configured to designate at least one target pixel of the plurality of pixels, wherein the plurality of read lines is configured to perform a reading process to read a specific signal from the at least one target pixel of the plurality of pixels; and
 a signal processing unit configured to perform a signal process based on the specific signal is included in a block, wherein specific width of the block is in a row direction of the matrix array, a height of the block is variable in a column direction of the matrix array, and the controller is further configured to:
 switch between a first mode and a second mode based on an instruction;
 designate, in the first mode, the at least one target pixel of the plurality of pixels after a thinning process on the plurality of pixels; and

29

designate, in the second mode, the at least one target pixel of the plurality of pixels without the thinning process.

2. The measurement device according to claim 1, wherein the controller is further configured to:

designate the at least one target pixel of the plurality of pixels with the block as a unit; and

control the thinning process such that a number of the at least one target pixel of the plurality of pixels included in the block in the first mode is equal to a number of the at least target pixel of the plurality of pixels included in the block in the second mode.

3. The measurement device according to claim 1, wherein the controller is further configured to designate the at least one target pixel of the plurality of pixels with the block as a unit.

4. The measurement device according to claim 1, wherein the controller is further configured to perform, in the first mode, the thinning process on the column direction of the matrix array.

5. The measurement device according to claim 1, wherein the controller is further configured to perform, in the first mode, the thinning process on the row direction of the matrix array.

6. The measurement device according to claim 1, wherein the controller is further configured to perform, in the first mode, the thinning process on the column direction of the matrix array and the row direction of the matrix array.

7. The measurement device according to claim 1, further comprising a plurality of signal lines, wherein each of the plurality of signal lines is configured to transmit a plurality of signals, the plurality of read lines is further configured to read the plurality of signals from the plurality of pixels, the plurality of pixels is in the light receiving unit along the column direction of the matrix array, each of the plurality of signal lines is connected to at least two pixels of the plurality of pixels, the at least two pixels of the plurality of pixels are at $(v-1)$ pixels along the column direction of the matrix array, v is an integer greater than or equal to two, and the controller is further configured to designate the thinning process at intervals at which the reading process is not performed from the at least two pixels of the plurality of or more pixels.

8. The measurement device according to claim 1, wherein the controller is further configured to stop power supply to a specific pixel of the plurality of pixels, and the specific pixel is not designated as the at least one target pixel of the plurality of pixels.

9. The measurement device according to claim 1, wherein the controller is further configured to switch between the first mode and the second mode based on the instruction from an outside.

10. A distance measuring device, comprising:

a light receiving unit configured to receive specific light from a light source, wherein

the light receiving unit includes a plurality of pixels in a target region in a matrix array, and

the plurality of pixels includes a plurality of read lines;

a controller configured to designate at least one target pixel of the plurality of pixels, wherein

the plurality of read lines is configured to perform a reading process to read a specific signal from the at least one target pixel of the plurality of pixels;

a time acquisition unit configured to measure a time period from a time of light emission to a time of light

30

reception based on a first pixel of the plurality of pixels is included in a block, wherein

the time of light emission corresponds to a time associated with emission of light by the light source,

the time of light reception corresponds to a time associated with the reception of the specific light,

a specific width of the block is in a row direction of the matrix array, and

a height of the block is variable in a column direction of the matrix array;

a generation unit configured to:

add a number of the measured time period in a specific time range based on the measured time period; and

generate, based on the addition of the number of the measured time period, a histogram associated with the block; and

a calculation unit configured to calculate a distance to an object based on the histogram, wherein the controller is further configured to:

switch between a first mode and a second mode based on an instruction;

designate, in the first mode, the at least one target pixel of the plurality of pixels after a thinning process on the plurality of pixels; and

designate, in the second mode, the at least one target pixel of the plurality of pixels without performing the thinning process.

11. The distance measuring device according to claim 10, wherein the controller is further configured to:

designate the at least one target pixel of the plurality of pixels with the block as a unit; and

control the thinning process such that a number of the at least one target pixel of the plurality of pixels included in the block in the first mode is equal to a number of the at least one target pixel of the plurality of pixels included in the block in the second mode.

12. The distance measuring device according to claim 10, wherein the controller is further configured to designate the at least one target pixel of the plurality of pixels with the block as a unit.

13. The distance measuring device according to claim 10, wherein the controller is further configured to perform, in the first mode, the thinning process on the column direction of the matrix array.

14. The distance measuring device according to claim 10, wherein the controller is further configured to perform, in the first mode, the thinning process on the row direction of the matrix array.

15. The distance measuring device according to claim 10, wherein the controller is further configured to perform, in the first mode, the thinning process on the column direction of the matrix array and the row direction of the matrix array.

16. The distance measuring device according to claim 10, further comprising a plurality of signal lines, wherein

each of the plurality of signal lines is configured to transmit a plurality of signals,

the plurality of read lines is further configured to read the plurality of signals from the plurality of pixels,

the plurality of pixels is in the light receiving unit along the column direction of the matrix array,

each of the plurality of signal lines is connected to at least two pixels of the plurality of pixels,

the at least two pixels of the plurality of pixels are at $(v-1)$ pixels along the column direction of the matrix array,

v is an integer greater than or equal to two, and

31

the controller is further configured to designate the thinning process at intervals at which the reading process is not performed from the at least two pixels of the plurality of pixels.

17. The distance measuring device according to claim **10**,
wherein

the controller is further configured to stop power supply to a second pixel of the plurality of pixels, and the second pixel is not designated as the at least one target pixel of the plurality of pixels.

18. The distance measuring device according to claim **10**,
wherein

the controller is further configured to switch between the first mode and the second mode based on the instruction from an outside.

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32